Probabilistic Methods

Lecture Notes

A clear route from counting and
events to random variables and data.

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Chapter 1

Introduction

These notes are an introduction to probabilistic thinking: the mathematical art of reasoning under uncertainty. Probability is not presented here as a collection of ready-made formulas, but as a language for describing experiments, incomplete information, variability, randomness, and data.

The purpose of the course is to teach how to build probabilistic models carefully. Before one computes a probability, expectation, variance, confidence interval, or approximation, one should understand what is random, what is being observed, what assumptions are being made, and what mathematical object represents the situation. In this sense, probability is not only a computational tool, but also a discipline of precise modelling.

The course is intended for readers who want to understand why probabilistic methods work, not only how to apply them. The emphasis is therefore placed on interpretation, model construction, and the connection between formal definitions and concrete situations. Calculations appear throughout the notes, but they are treated as consequences of a well-defined model rather than as isolated procedures.

1.1 How to Use These Notes

These notes are meant to be read before problem classes. Their role is not only to present the material, but also to show how mathematical argumentation is built: how one starts from a situation, identifies the relevant objects, formulates assumptions, chooses notation, develops a line of reasoning, and only then performs computations. A problem class is therefore not the place where the basic language of the topic is introduced for the first time; it is the place where that language is used, tested, discussed, and refined through problems.

Students are expected to read the relevant chapter in advance and to treat the text as a model of reasoning. The important question is not only "What is the answer?" but also "How is the solution argued?" While reading, students should pay attention to the order of the argument, the meaning of each definition, the reason for each assumption, and the way examples are connected to general concepts. The aim is to learn how to construct a probabilistic explanation, not merely how to recognize a formula that seems to fit a task.

Problem lists will accompany the chapters and will be used during classes. They are meant to be solved after the corresponding text has been read. The chapter gives the language, concepts, and patterns of argumentation needed to approach the problems responsibly. Without this preparation, a solution may look complete on paper while still being mathematically empty: it may contain formulas, calculations, or generated text, but no clear understanding of what model is being used or why the reasoning is valid.

Artificial intelligence tools may be useful for practice, checking calculations, generating additional examples, or asking for alternative explanations. Students are encouraged to use such tools responsibly when they support learning. However, AI-generated text cannot replace the student's own reading and understanding. A student must be able to read, question, correct, and explain any solution they present. If one cannot explain the experiment, the elementary outcomes, the assumptions, the chosen model, and the steps of the argument, then the work has not yet been done.

These notes also help students learn what to ask for. To use problem lists, textbooks, or AI tools well, one must first know what matters: which object is being modelled, which assumptions are decisive, which definitions are involved, and what kind of justification is expected. The PDF is therefore an essential part of the course: it teaches the material, but it also teaches how to read mathematics, how to demand proper explanations, and how to recognize whether an answer contains real reasoning.

The expected preparation for classes is conceptual as well as computational. After reading a chapter, a student should be able to summarize its main ideas, explain the definitions in their own words, reconstruct the key examples, identify the assumptions in a probabilistic model, follow and reproduce the main arguments, and answer questions that test understanding of the text. The notes are deliberately written with full explanations and detailed argumentation so that this preparation is possible.

1.2 Structure of the Course

These lecture notes are organized as a gradual construction of probabilistic thinking. The aim is not to begin with formulas and then search for examples, but to develop the mathematical language needed to describe uncertainty in a precise way. We start from the problem of counting and representing possible outcomes, then move toward events, probability, random variables, probability laws, empirical data, sampling, limit theorems, and statistical inference.

The course begins with combinatorics because many probabilistic models depend on the ability to count possible outcomes correctly. At this stage probability itself is not yet needed. What is needed is the more elementary skill of recognizing what kind of object an outcome is: a sequence, a selection, an arrangement, a choice with or without repetition, or an object whose elements may or may not be distinguishable. This first step builds the counting language that later allows us to describe sample spaces and favorable outcomes with precision.

The second stage has a special conceptual role. It is devoted to discovering the formal language of probability theory. The purpose is to show how a mathematical theory can be built from careful thinking, analysis, and abstraction. We examine how statements about a random situation select sets of outcomes, how these sets become events, how logical operations correspond to set operations, and why a numerical theory of probability should be placed on such a structure. This part is not primarily computational; it is meant to teach how probabilistic language is formed.

After this conceptual preparation, the notes turn to sample spaces and elementary probability. Several canonical examples are developed from the beginning: one first understands the experiment, decides what should count as an elementary outcome, constructs the sample space, defines events, and only then assigns probabilities. Different representations of the same structure, such as lists, tables, trees, and symbolic notation, are used to show that a sample space is an abstract mathematical model rather than a collection of observed data.

The next step is conditional probability. Here the central idea is that information changes the space in which we reason. Conditional probability is introduced by asking what it means to restrict attention to the cases in which some event has already occurred. From this point of view, the formula for conditional probability becomes natural, and it leads to independence, dependence, partitions, the law of total probability, and Bayes' theorem. Bayes' theorem is treated as a central tool for reversing conditional reasoning and for understanding situations where intuition is easily misled.

The course then introduces random variables through the discrete case. A random variable appears as a function that extracts a numerical quantity from an elementary outcome: the number of successes, the number of errors, the number of arrivals, a payoff, or another measurable feature of the experiment. This leads to the support of a discrete random variable, its probability mass function, its cumulative distribution function, expectation, variance, moments, and interpretation. Classical discrete laws are presented as models arising from specific mechanisms, not as isolated formulas.

The continuous case is then developed as a parallel but conceptually more demanding part of the theory. For a continuous random variable, probability is no longer concentrated at individual points, so probabilities of intervals become the fundamental objects. This leads to density functions, cumulative distribution functions, integrals, continuous expectations, variances, moments, quantiles, and the interpretation of probability as area under a density. Important continuous models, including the uniform, exponential, normal, gamma, and beta distributions, are introduced with attention to their meaning and use.

Once theoretical probability laws have been introduced, the course moves to empirical distributions and descriptive statistics. The same ideas now appear in data: frequency tables, histograms, empirical cumulative distribution functions, means, medians, variances, standard deviations, quartiles, interquartile ranges, and outliers. This stage is meant to clarify the difference between an idealized probability law and a finite dataset, and also to show how empirical summaries mirror the theoretical quantities defined earlier.

The following stage adds another level of randomness by studying sampling and distributions of statistics. A sample mean, sample proportion, sample variance, median, or other statistic is not merely a number; it is itself a random variable because it depends on the random sample. This perspective prepares the ground for understanding sampling distributions and for seeing why repeated sampling creates a new layer of probabilistic structure.

The final stage of the present course outline is devoted to limit theorems and statistical inference. Once empirical summaries and sampling distributions are understood, we can ask what regularities appear as the sample size grows or as experiments are repeated many times. This leads to the law of large numbers, the central limit theorem, normal approximations, standard errors, confidence intervals, and the basic logic of statistical inference. The goal is to show how probability theory makes inference from data possible: it allows us to estimate unknown quantities, measure uncertainty, and justify conclusions mathematically.

Chapter 2

Elements of Combinatorics

2.1 Counting begins with elementary outcomes

Combinatorics is often presented as a list of formulas. This is misleading. Before a formula can be used, we must know what one elementary outcome is. A physical situation does not determine a mathematical space of possibilities by itself. The space of possibilities depends on what is recorded.

Suppose that three balls are drawn from a box. If the balls are labelled and the order of drawing is recorded, then an outcome is a sequence of labelled objects. If only the selected labelled balls are recorded, then an outcome is a set. If only the colors are recorded, then labelled balls of the same color may become indistinguishable. If only the number of red balls is recorded, then the outcome is a numerical summary of the draw rather than the full draw itself.

Thus the first question in a counting problem is not

Which formula should be used?

The first question is

What is one outcome?

Definition

An **elementary outcome** is a complete description of one possible result of a situation, according to the recording rule chosen for the model.

The phrase "according to the recording rule" is essential. The same physical situation may produce different mathematical spaces of possibilities if different information is recorded.

Example

A box contains three balls labelled R_1, R_2, B_1. Two balls are drawn without replacement. If order and labels are recorded, then examples of outcomes are

$$(R_1, B_1), \qquad (B_1, R_1), \qquad (R_1, R_2).$$

If only the set of labelled balls is recorded, then examples of outcomes are

$$\{R_1, B_1\}, \qquad \{R_1, R_2\}.$$

If only the sequence of colors is recorded, then examples of outcomes are

$$(R, B), \qquad (B, R), \qquad (R, R).$$

If only the number of red balls is recorded, then the possible recorded values are

$$0, 1, 2.$$

These are not four different answers to the same counting question. They are four different mathematical models of the same physical situation.

Remark

A counting error often begins before any arithmetic is done. It begins when the outcome type is chosen incorrectly.

2.2 Recording rules and spaces of possibilities

A recording rule tells us which differences between physical results are mathematically visible. Some differences are recorded; others are ignored. Once the recording rule is fixed, the space of possibilities is the set of all possible recorded outcomes.

Definition

A **recording rule** is a decision about what information is included in the mathematical outcome and what information is ignored.

For finite situations, the construction usually follows this pattern:

$$\text{physical situation} \longrightarrow \text{recording rule} \longrightarrow \text{elementary outcome} \longrightarrow \text{space of possibilities.}$$

The following table shows common outcome types.

Recording rule	Outcome type	Typical notation
Order is recorded	sequence	$(a_1, a_2, \ldots, a_k)$
Only membership is recorded	set	$\{a_1, a_2, \ldots, a_k\}$
All objects are arranged in a row	linear arrangement	$(a_{\sigma(1)}, \ldots, a_{\sigma(n)})$
Objects are arranged on a circle	circular arrangement	relative order
Only numbers of types are recorded	count vector	$(n_1, n_2, \ldots, n_r)$
Only a numerical summary is recorded	summary value	x

2.3 Representing the space of possibilities

Counting is easier when the space of possibilities is represented in a form that matches the structure of the problem. A representation is not decoration. It is part of the reasoning, because it makes visible which outcomes are different, which descriptions are equivalent, and which construction steps are being used.

Different situations call for different representations:

- a list is useful when the space is small and every outcome can be written explicitly;
- a table is useful when two choices are made and rows and columns organize the possibilities;
- a tree is useful when an outcome is built through successive stages;
- a set notation is useful when only membership matters;
- a tuple notation is useful when positions or order matter;
- a count vector is useful when only the numbers of objects of each type matter.

Example

If two letters are chosen from $\{A, B, C\}$ and order is recorded with repetition allowed, the space can be represented as a table:

	A	B	C
A	(A, A)	(A, B)	(A, C)
B	(B, A)	(B, B)	(B, C)
C	(C, A)	(C, B)	(C, C)

The table shows two things at once: there are 3 choices for the first position and 3 choices for the second position, and the ordered pairs (A, B) and (B, A) are different outcomes.

Example

If two letters are chosen from $\{A, B, C\}$ and only the selected set is recorded, then

$$\{A, B\} = \{B, A\}.$$

A table of ordered pairs would still be possible as an auxiliary representation, but it would overdescribe the outcomes. The actual outcomes are sets:

$$\{A, B\}, \qquad \{A, C\}, \qquad \{B, C\}.$$

The change of representation reflects a change of model.

A useful representation should answer the question: what differences are visible in this model? If the representation hides an important difference, the count may be too small. If it records irrelevant differences, the count may be too large.

Example

A committee of three people is chosen from ten people. If the committee is recorded only as a group, then

$$\{A, B, C\} = \{C, A, B\}.$$

The order of listing the names is irrelevant. The outcome is a set.
If the same three people are assigned the roles of chair, secretary, and treasurer, then

$$(A, B, C) \neq (C, A, B).$$

The outcome is now an ordered assignment of roles.

The two situations may look similar in everyday language: in both cases three people are selected. But mathematically they are different because the recording rules are different.

2.4 Diagnostic questions before counting

Before using a formula, answer the following questions.

1. What is one elementary outcome?
2. Is the outcome a sequence, a set, an arrangement, or a count vector?
3. Does order matter?

4. Are all objects used, or only some of them?
5. Can the same object be used more than once?
6. Are the objects distinguishable or indistinguishable?
7. Are we first counting labelled descriptions and then identifying descriptions that represent the same outcome?

These questions are more important than the names of formulas. The names are only labels attached to structures that have already been recognized.

Remark

Do not begin a problem by asking which formula you remember. Begin by asking what kind of outcome is being constructed.

2.5 The multiplication principle as a construction rule

Many counting problems can be solved by describing how an outcome is constructed step by step.

Theorem

Multiplication principle. Suppose a construction is performed in k successive stages. If there are m_1 choices for the first stage, m_2 choices for the second stage, and so on up to m_k choices for the k-th stage, then the number of possible complete constructions is

$$m_1 m_2 \cdots m_k.$$

The multiplication principle is not only a formula. It is a way of reading an outcome as a construction process.

Example

A four-digit PIN code is formed from the digits $0, 1, \ldots, 9$, and digits may repeat. One outcome is a sequence such as

$$(0, 7, 7, 3).$$

There are 10 choices for each position, hence

$$10 \cdot 10 \cdot 10 \cdot 10 = 10^4$$

possible PIN codes.
The reason is not that the formula n^k was memorized. The reason is that a PIN code is a sequence of four independent choices from a set of ten symbols.

Example

A three-letter code is formed from the letters A, B, C, D, and letters may not repeat. One outcome is an ordered triple, for example

$$(A, D, B).$$

There are 4 choices for the first position, 3 for the second, and 2 for the third. Therefore the number of codes is

$$4 \cdot 3 \cdot 2 = 24.$$

When the same number of choices is available at every stage, powers appear. When the number of choices decreases, falling products appear. When order is later ignored, division appears. Most elementary combinatorial formulas can be understood from these three ideas.

2.6 Sequences with repetition

A sequence with repetition is an ordered list in which each position is filled separately from the same set of symbols.

Definition

Let S be a set with n elements. A **sequence of length k with repetition** over S is an ordered k-tuple

$$(a_1, a_2, \ldots, a_k),$$

where each $a_i \in S$. The same element may occur many times.

The word "ordered" means that positions matter. The first position, the second position, and the third position are different places in the description. The phrase "with repetition" means that choosing one symbol at one position does not remove it from later positions.

Now we derive the count. To construct one sequence,

- choose a_1: there are n choices;
- choose a_2: there are again n choices, because repetition is allowed;
- continue in the same way up to a_k: there are n choices at each position.

By the multiplication principle, the number of complete sequences is

$$\underbrace{n \cdot n \cdots n}_{k \text{ factors}} = n^k.$$

Thus the formula n^k is not a separate rule to memorize. It is the multiplication principle applied to k ordered positions, each with the same n choices.

This model is used when:

- order is recorded,
- each position is chosen separately,
- repetition is allowed,
- the outcome is a full sequence.

Typical examples include PIN codes, repeated coin tosses, repeated die rolls, strings over an alphabet, and repeated selections with replacement.

Example

The space of possibilities for three tosses of a coin is

$$\{H, T\}^3.$$

It has

$$2^3 = 8$$

elementary outcomes. Each outcome is a sequence, for example

$$(H, T, H).$$

The outcomes (H, T, H) and (T, H, H) are different because the order of positions is recorded.
A tree representation shows the same count visually. At the first toss there are 2 branches, from each of these there are 2 branches for the second toss, and from each of those there are 2 branches for the third toss. The tree has

$$2 \cdot 2 \cdot 2 = 8$$

terminal branches, one for each complete sequence.

2.7 Ordered selections without repetition

Sometimes only some objects are chosen, but the order or position of the chosen objects matters.

Definition

An **ordered selection without repetition** of length k from n distinct objects is an ordered list of k different objects chosen from the n available objects.

To derive the count, construct the ordered list position by position. There are n choices for the first position. Once the first object has been used, it cannot be used again, so there are $n - 1$ choices for the second position. After two objects have been used, there are $n - 2$ choices for the third position. Continuing in this way, the k-th position has

$$n - (k - 1) = n - k + 1$$

choices. Therefore the number of ordered selections without repetition is

$$P(n, k) = n(n - 1)(n - 2) \cdots (n - k + 1).$$

This product can also be written using factorials. Since

$$n! = n(n - 1) \cdots (n - k + 1)(n - k)(n - k - 1) \cdots 1,$$

dividing by $(n - k)!$ removes the unused tail:

$$P(n, k) = \frac{n!}{(n - k)!}.$$

This model is used when:

- only some objects are selected,
- order or position matters,
- no object may be used twice.

Example

Gold, silver, and bronze medals are assigned among 12 runners. An outcome is not a set of three runners. It is an ordered triple

$$(\text{gold}, \text{silver}, \text{bronze}).$$

There are 12 choices for gold, then 11 choices for silver, then 10 choices for bronze. Hence the number of possible assignments is

$$12 \cdot 11 \cdot 10 = P(12, 3).$$

In some European terminology, ordered selections are called variations. In these notes we use the name ordered selections without repetition, or k-permutations, because it describes directly what the outcome is.

2.8 Arrangements of all objects

A permutation is an arrangement of all objects in a row.

Definition

A **permutation** of n distinct objects is a linear arrangement of all n objects.

This is the special case of an ordered selection without repetition in which all objects are used. Thus $k = n$, and

$$P(n, n) = n(n-1)(n-2)\cdots 2 \cdot 1 = n!.$$

So there are

$$n!$$

permutations of n distinct objects.

Example

If 7 students are arranged in a line, then one outcome is a complete ordered list of all students. The number of arrangements is

$$7!.$$

This is not because 7! is a name attached to the word "arrangement". It is because there are 7 choices for the first position, 6 for the second, 5 for the third, and so on until all positions are filled.

Permutations are used when:

- all objects are used,
- order matters,
- the objects are distinguishable,
- the arrangement is linear.

2.9 Circular arrangements

A circular arrangement is different from a linear arrangement because rotating the whole arrangement does not create a new relative seating order.

Example

Suppose four people A, B, C, D sit around a round table. The linear lists

$$(A, B, C, D), \qquad (B, C, D, A), \qquad (C, D, A, B), \qquad (D, A, B, C)$$

represent the same circular arrangement if only relative positions matter.

For n distinct objects placed around a circle, with rotations identified, the number of circular arrangements is

$$(n-1)!.$$

One way to see this is to fix one object as a reference point. This does not reduce any real choice; it only chooses one representative from each group of rotated descriptions. Once the reference object is fixed, the remaining $n-1$ objects can be placed around it in a linear order:

$$(n-1)(n-2)\cdots 2\cdot 1 = (n-1)!.$$

Remark

A circular arrangement requires a modelling decision. If seats are numbered or if there is a distinguished position, then rotations may become different outcomes and the linear count may be appropriate.

2.10 Unordered selections: combinations

A combination is an unordered selection.

Definition

A **combination** of size k from an n-element set is a subset with k elements.

The number of combinations is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

This formula can be derived from ordered selections. There are

$$P(n,k) = \frac{n!}{(n-k)!}$$

ordered selections of length k. But each unordered selection of k objects can be ordered in $k!$ ways. Therefore

$$\binom{n}{k} = \frac{P(n,k)}{k!} = \frac{n!}{k!(n-k)!}.$$

Combinations are used when:

- only membership is recorded,
- order is ignored,

- each object may be chosen at most once,
- the outcome is a set.

Example

A five-card hand from a standard deck is a set of five cards, not a sequence of five draws, if the final hand is all that is recorded. The number of possible hands is

$$\binom{52}{5}.$$

If the order of drawing is recorded, the correct count is instead

$$P(52,5) = 52 \cdot 51 \cdot 50 \cdot 49 \cdot 48.$$

These are different sample spaces.

2.11 Overcounting and equivalent descriptions

Many combinatorial formulas arise from a two-step idea:

1. Count descriptions that are easy to construct.
2. Divide by the number of descriptions representing the same outcome.

This is the idea of overcounting.

Definition

Overcounting occurs when the same intended outcome is counted more than once because it has several different descriptions.

Overcounting is not a mistake if it is controlled. It becomes a method when we know exactly how many times each outcome has been counted.

Example

Suppose we want to choose a committee of three people from n people. The order of names in the committee does not matter.
It is easier first to count ordered selections:

$$n(n-1)(n-2).$$

But each committee of three people has been counted $3!$ times, because the same three people can be listed in $3!$ different orders. Therefore the number of committees is

$$\frac{n(n-1)(n-2)}{3!}.$$

The essential idea is not division itself. The essential idea is identifying which descriptions should be treated as the same outcome.

2.12 Distinguishable and indistinguishable objects

Objects are distinguishable if the model treats them as separate individuals. Objects are indistinguishable if only their type matters.

Example

The objects

$$R_1, R_2, R_3$$

are three distinguishable red balls. The symbols

$$R, R, R$$

represent three indistinguishable red balls of the same type.

A physical object may be distinguishable in one model and indistinguishable in another. This depends on the recording rule.

Example

An urn contains red and blue balls. If each ball has a label, and labels are recorded, then two red balls are different outcomes. If only colors are recorded, then the red balls are treated as indistinguishable.

This distinction is crucial in probability. If the physical random mechanism selects labelled balls uniformly, but the sample space records only colors, then the color outcomes may not be equally likely.

2.13 Permutations with repeated elements

When some objects are indistinguishable, a linear arrangement may have several labelled descriptions that represent the same visible arrangement.

Suppose we arrange n objects, where there are n_1 objects of type 1, n_2 objects of type 2, and so on, with

$$n_1 + n_2 + \cdots + n_r = n.$$

The number of distinct visible arrangements is

$$\frac{n!}{n_1!n_2!\cdots n_r!}.$$

Example

The word BANANA contains six letters: three A's, two N's, and one B. If all six positions are arranged, then the number of distinct words is

$$\frac{6!}{3!2!1!}.$$

The division appears because permuting the three A's among themselves does not change the visible word, and neither does permuting the two N's.

This formula should be read as an overcounting correction:

$$\text{labelled arrangements} \quad \longrightarrow \quad \text{visible arrangements.}$$

2.14 Combinations with repetition

Sometimes we choose k objects from n types, and the same type may be chosen several times. The order of the chosen objects is ignored. The outcome is then not a sequence of choices, but a vector of counts.

Definition

A **combination with repetition** of size k from n types is a choice of non-negative integers

$$(x_1, x_2, \ldots, x_n)$$

such that

$$x_1 + x_2 + \cdots + x_n = k.$$

Here x_i records how many objects of type i are chosen.

The important modelling decision is that the positions of the chosen objects are not recorded. For example, choosing two objects of type 1, none of type 2, and three of type 3 is recorded as

$$(2, 0, 3),$$

not as an ordered sequence of five choices.

The stars and bars representation

We derive the formula by representing each count vector visually. Write k stars to represent the chosen objects and insert $n-1$ bars to divide the stars into n groups. The number of stars before the first bar is x_1, the number of stars between the first and second bar is x_2, and so on, until the number of stars after the last bar is x_n.

For example, if $n = 3$ and $k = 5$, the count vector

$$(2, 0, 3)$$

is represented by

$$* * \,|\,|\, * * * \,.$$

There are two stars before the first bar, no stars between the two bars, and three stars after the second bar. This represents

$$x_1 = 2, \qquad x_2 = 0, \qquad x_3 = 3.$$

Conversely, every arrangement of k stars and $n-1$ bars determines exactly one count vector. For instance,

$$|\, * * * \,|\, * *$$

represents

$$(0, 3, 2).$$

This is a one-to-one correspondence:

$$\text{count vectors } (x_1, \ldots, x_n) \text{ with } x_1 + \cdots + x_n = k \longleftrightarrow \text{arrangements of } k \text{ stars and } n-1 \text{ bars.}$$

Therefore the counting problem becomes a simpler arrangement problem. We have

$$k + (n-1) = n + k - 1$$

total positions. To form an arrangement, we only need to decide which k of these positions contain stars; the remaining $n-1$ positions will contain bars. Hence the number of arrangements is

$$\binom{n+k-1}{k}.$$

Equivalently, we may choose the $n-1$ positions of the bars, giving

$$\binom{n+k-1}{n-1}.$$

Thus

$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$

Theorem

The number of combinations with repetition of size k from n types is

$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$

Example

How many ways are there to choose 5 donuts from 3 flavors if only the numbers of each flavor matter?
An outcome is a count vector

$$(x_1, x_2, x_3), \qquad x_1 + x_2 + x_3 = 5.$$

The stars and bars representation uses 5 stars and 2 bars, so there are 7 positions in total. We may choose the 5 positions of the stars:

$$\binom{7}{5},$$

or the 2 positions of the bars:

$$\binom{7}{2}.$$

Thus the number of possible choices is

$$\binom{3+5-1}{5} = \binom{7}{5} = \binom{7}{2}.$$

Remark

Combinations with repetition do not describe ordered repeated choices. If order matters, the model is a sequence with repetition and the count is n^k. If order is ignored and only counts of types matter, the model is combinations with repetition.

2.15 Recognizing the correct counting model

The following diagnostic table summarizes the main finite counting structures.

Outcome type	Order?	Repetition?	Count
sequence with repetition	yes	yes	n^k
ordered selection	yes	no	$P(n,k) = \dfrac{n!}{(n-k)!}$
permutation	yes	no, all used	$n!$
circular arrangement	relative order	no, all used	$(n-1)!$
combination	no	no	$\binom{n}{k}$
combination with repetition	no	yes	$\binom{n+k-1}{k}$
multiset permutation	positions yes	identical objects	$\dfrac{n!}{n_1! \cdots n_r!}$

The table should be used only after the outcome type has been identified. It is a summary, not a substitute for modelling.

Remark

A formula is a compressed argument. If the argument does not match the situation, the formula gives the wrong count even when the arithmetic is correct.

2.16 Common traps

Trap 1: confusing a sequence with a set

The ordered outcomes

$$(A, B, C) \quad \text{and} \quad (C, B, A)$$

are different sequences but the same set:

$$\{A, B, C\} = \{C, B, A\}.$$

Use sequences when order is recorded. Use sets when only membership is recorded.

Trap 2: treating indistinguishable objects as distinguishable

The letters in BANANA are not six distinct letters. The three A's are indistinguishable, and the two N's are indistinguishable. Therefore 6! overcounts the number of visible arrangements.

Trap 3: using combinations when roles matter

A committee is a set. A medal assignment is an ordered assignment. A group of three athletes and a gold-silver-bronze result are not the same kind of outcome.

Trap 4: using a representation that records too much

Sometimes an auxiliary representation is more detailed than the intended outcome. For example, ordered lists may be useful for deriving the number of committees, but the committee itself is not an ordered list. If the extra descriptions are not corrected for, the count is too large.

Trap 5: using a representation that records too little

Sometimes a representation hides distinctions that the model requires. If positions, roles, labels, or order are part of the outcome, then a representation that ignores them gives too small a count. A good representation must preserve exactly the distinctions that the recording rule declares visible.

2.17 What combinatorics teaches here

The role of this chapter is not to memorize more formulas. Its role is to teach how to build and count finite spaces of possibilities. A formula is useful only after the underlying structure has been recognized.

The general workflow is

$$\text{situation} \longrightarrow \text{recording rule} \longrightarrow \text{elementary outcome} \longrightarrow \text{space of possibilities} \longrightarrow \text{count.}$$

Each arrow in this workflow represents a mathematical decision. The same everyday description may lead to different counts if the recording rule changes. The same physical objects may be treated as distinguishable or indistinguishable depending on what is visible in the model. The same collection may be represented as a list, a set, an arrangement, or a count vector.

Good counting therefore has two parts. First, one must construct the correct model. Second, one must justify the count by a construction argument, an overcounting correction, or a representation such as a table, a tree, or a stars-and-bars diagram.

2.18 Final checklist

When solving a counting problem, write down the following before doing arithmetic:

1. What is the physical situation?
2. What information is recorded?
3. What is one elementary outcome?
4. What is the space of possibilities?
5. How can this space be represented: as a list, table, tree, set, tuple, arrangement, or count vector?
6. Are outcomes sequences, sets, arrangements, or count vectors?
7. Does the representation record exactly the distinctions required by the model?
8. Which construction, overcounting correction, or bijection gives the count?
9. What does the final number mean in the original situation?

This checklist is the main lesson of the chapter. The formulas are useful, but they are useful only when the underlying structure has been discovered first and the count has been justified.

Chapter 3

Discovering Formalism

3.1 Why formalism has to be discovered

The previous chapter taught a first lesson: before one counts, one must know what is being counted. The present chapter begins from a closely related but deeper observation. Before one assigns any numerical value to uncertainty, one must know what kind of object a statement about a situation selects.

Mathematics often begins when ordinary language becomes too elastic. In ordinary language we can say things such as

$$\text{the result is favourable,} \qquad \text{at least one die shows 6,}$$

$$\text{the first toss is heads,} \qquad \text{it rains on Wednesday or Friday.}$$

These statements are meaningful, but they are not yet mathematical objects. They have grammar, but not yet structure. To reason with them reliably, we must translate them into something that can be compared, combined, negated, represented, and eventually measured.

The central discovery of this chapter is that a statement about a situation acts as a test on elementary outcomes. An elementary outcome either satisfies the statement or it does not. Therefore a statement selects a set of elementary outcomes. This selected set is what probability theory calls an event.

Remark

The formal language is not introduced to make the subject look more abstract. It is introduced because without it we cannot reliably say what is being discussed.

At this stage, we are not primarily interested in numerical answers. We are interested in the birth of a language. We want to see how the following chain of ideas appears naturally:

$$\text{situation} \longrightarrow \text{elementary outcomes} \longrightarrow \text{statements about outcomes} \longrightarrow \text{sets of outcomes} \longrightarrow \text{operations on set}$$

Only after this language exists can numerical probability be introduced responsibly.

3.2 A first universe of discourse

Every formal discussion needs a universe in which its statements are interpreted. In our setting, that universe is the set of all elementary outcomes under consideration.

Definition

The **sample space** Ω is the set of elementary outcomes that are possible under the recording rule chosen for the situation.

In this chapter, Ω should be read as a universe of discourse. It tells us what the word "outcome" means in the current model. A statement is not evaluated in the air. It is evaluated relative to Ω.

Example

Consider two coin tosses, with order recorded. The sample space is

$$\Omega = \{HH, HT, TH, TT\}.$$

The statement "exactly one head occurs" is true for HT and TH, but false for HH and TT. Thus the statement selects the set

$$\{HT, TH\}.$$

The selected set is the mathematical object corresponding to the statement.

The same phrase may select a different set if the sample space changes. For example, if only the number of heads is recorded, then the sample space is

$$\Omega' = \{0, 1, 2\}.$$

In that model, the statement "exactly one head occurs" selects

$$\{1\}.$$

The ordinary sentence is similar, but the formal object is different because the underlying space is different.

Remark

Formalism forces us to keep track of the level at which we are speaking. Are we describing the full ordered outcome, a set of visible outcomes, or only a numerical summary? The answer changes the mathematical object.

This is the methodological point of the chapter. We do not begin with symbols because symbols are fashionable. We begin with a question: what must be preserved so that reasoning remains honest? The symbols appear as a disciplined way to preserve exactly that structure.

3.3 Statements become events

Let Ω be fixed. A statement about the situation becomes mathematical when we can say exactly for which elementary outcomes it is true. The set of all such outcomes is called an event.

Definition

An **event** is a subset of the sample space Ω. If $A \subseteq \Omega$, then A consists of precisely those elementary outcomes for which a certain statement is true.

This definition may look simple, but it is a major conceptual move. It tells us that statements and sets can play the same role once Ω has been fixed:

$$\text{statement about the situation} \quad \longleftrightarrow \quad \text{subset of } \Omega.$$

The statement gives meaning. The subset gives structure.

Example

Suppose two dice are rolled and order is recorded. Then

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}.$$

The statement

the sum is 7

selects the event

$$A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

The statement

the first die is greater than the second

selects the event

$$B = \{(i, j) \in \Omega : i > j\}.$$

The notation $B = \{(i, j) \in \Omega : i > j\}$ is not a shortcut for avoiding thought. It records the rule by which membership in the event is decided.

3.4 Representing events as regions

A set of outcomes can be written as a list, but lists are not always the best representation. When Ω has a visible structure, an event should often be represented as a region inside that structure. This is not merely a visual aid. It is a way of seeing relations.

For two dice, the sample space is naturally represented as a 6×6 table:

	1	2	3	4	5	6
1	.	.	.	.	.	.
2	.	.	.	.	.	.
3	.	.	.	.	.	.
4	.	.	.	.	.	.
5	.	.	.	.	.	.
6	.	.	.	.	.	.

Each cell represents one ordered pair (i, j). The row records the first die, and the column records the second die.

The event $B = \{(i, j) : i > j\}$ occupies the cells below the diagonal:

	1	2	3	4	5	6
1	.	.	.	.	.	.
2	X	.	.	.	.	.
3	X	X	.	.	.	.
4	X	X	X	.	.	.
5	X	X	X	X	.	.
6	X	X	X	X	X	.

The diagonal itself corresponds to $i = j$. The region above the diagonal corresponds to $i < j$. The geometry of the table reveals the structure of the statement.

Remark

A representation is good when it preserves the distinctions relevant to the statement and makes relations visible. A poor representation may hide the structure that the problem is trying to teach.

3.5 Reading a region as a statement

The translation works in both directions. A verbal statement selects a set, but a marked set also asks to be interpreted as a statement.

Example

In the two-dice table, suppose the marked cells are exactly the diagonal:

	1	2	3	4	5	6
1	X	.	.	.	.	.
2	.	X	.	.	.	.
3	.	.	X	.	.	.
4	.	.	.	X	.	.
5	.	.	.	.	X	.
6	.	.	.	.	.	X

This region can be described in several equivalent ways:

the two dice show the same number,

the first result equals the second result,

$$\{(i, j) \in \Omega : i = j\}.$$

The ability to move between these descriptions is part of understanding the formalism.

Students often think that formal notation is a translation from language into symbols. That is only half of the work. One must also be able to read symbols and diagrams back into language. A set without interpretation is mute; a sentence without a set is vague. Mathematics becomes powerful when both sides illuminate each other.

3.6 Logical operations become set operations

Once statements have become events, logical operations on statements become set operations on events. This is one of the most important structural discoveries in elementary probability theory. The words

and, or, not

are not informal decorations. They determine precise operations on subsets of Ω.

Let $A \subseteq \Omega$ and $B \subseteq \Omega$ be events.

Language	Set operation	Meaning
A and B	$A \cap B$	outcomes satisfying both statements
A or B	$A \cup B$	outcomes satisfying at least one statement
not A	$A^c = \Omega \setminus A$	outcomes not satisfying A
A but not B	$A \setminus B$	outcomes satisfying A and not B

The table is not a dictionary of symbols to memorize. It expresses a deeper fact: logical structure and set structure are the same structure viewed in two languages.

Definition

For events $A, B \subseteq \Omega$:

$$A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\},$$

$$A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\},$$

$$A^c = \{\omega \in \Omega : \omega \notin A\}.$$

3.7 Compound statements in a two-dice universe

Consider again two dice, with

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}.$$

Define three events:

$$A = \{(i, j) : i + j = 7\},$$

$$B = \{(i, j) : i > j\},$$

$$C = \{(i, j) : i = 6 \text{ or } j = 6\}.$$

Now each compound statement can be built systematically.

The statement

the sum is 7 or at least one die shows 6

corresponds to

$$A \cup C.$$

The statement

the sum is 7 and at least one die shows 6

corresponds to

$$A \cap C.$$

The statement

the sum is 7 but the first die is not greater than the second

corresponds to

$$A \cap B^c.$$

The statement

at least one die shows 6 but the sum is not 7

corresponds to

$$C \setminus A = C \cap A^c.$$

This is the beginning of an algebra of events. We are not merely naming sets. We are learning to calculate with statements.

Remark

The purpose of the notation is not to replace thought. The purpose is to make the structure of thought inspectable. If a student writes $A \cap B^c$, they should also be able to say what it means in words and mark the corresponding region in the sample space.

3.8 Why "or" needs care

In everyday speech, the word "or" is sometimes ambiguous. In mathematics, the union $A \cup B$ means inclusive or: an outcome belongs to $A \cup B$ if it belongs to A, or to B, or to both.

This matters because the overlap is not automatically empty. If A and B overlap, then the union is not obtained by placing two disjoint blocks side by side. The intersection must be seen.

Example

Let

$$A = \{(i, j) : i + j = 7\}, \qquad C = \{(i, j) : i = 6 \text{ or } j = 6\}.$$

The outcome $(1, 6)$ belongs to both A and C. The outcome $(6, 1)$ also belongs to both. Thus

$$A \cap C = \{(1, 6), (6, 1)\}.$$

The phrase "sum is 7 or at least one die shows 6" includes these outcomes once, not twice. The union is a set of outcomes, not a list of reasons.

This is a methodological lesson. A sentence may have several reasons for being true, but an elementary outcome either belongs to the event or it does not. Formalism prevents double counting at the level of meaning before any numerical computation begins.

3.9 Complementation and the boundary of the universe

The complement A^c is always taken relative to the current sample space Ω. This is why Ω must be fixed before complements are meaningful.

Example

If two dice are rolled and

$$A = \{(i, j) : i + j = 7\},$$

then

$$A^c = \{(i, j) \in \Omega : i + j \neq 7\}.$$

The complement is not an abstract collection of all things in the world for which the sum is not 7. It is the collection of outcomes inside the two-dice sample space for which the sum is not 7.

The complement teaches a philosophical lesson hidden inside a technical definition: negation is always negation within a universe. To say "not A" mathematically, one must first say what counts as a possible case.

3.10 From events to observed frequencies

So far, an event is only a set of outcomes. This is already a substantial achievement: we can now state clearly what a sentence means. But probability theory eventually wants to do something more. It wants to attach numbers to events.

The first honest source of such numbers is observation. If an experiment is repeated many times, and we record how often each outcome or event occurs, then we obtain observed frequencies. These frequencies are not yet theoretical probabilities. They are empirical measurements. Nevertheless, they reveal the shape that a future theory of probability should have.

Suppose a die is rolled 1000 times. The elementary outcomes are

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Assume the recorded numbers are

$$n(\{1\}) = 168, \quad n(\{2\}) = 154, \quad n(\{3\}) = 181,$$

$$n(\{4\}) = 167, \quad n(\{5\}) = 160, \quad n(\{6\}) = 170.$$

For an event $A \subseteq \Omega$, define its observed frequency by

$$f(A) = \frac{n(A)}{1000},$$

where $n(A)$ is the number of recorded trials whose outcome belongs to A.

Example

Let

$$A = \{2, 4, 6\}.$$

Then A occurs whenever the recorded die result is 2, 4, or 6. Therefore

$$n(A) = n(\{2\}) + n(\{4\}) + n(\{6\}) = 154 + 167 + 170 = 491,$$

and hence

$$f(A) = \frac{491}{1000} = 0.491.$$

The addition is justified because the elementary outcomes 2, 4, and 6 are disjoint possibilities. A single roll cannot be both 2 and 4.

This example shows a transition. We did not begin with a formula for probability. We began with a set A, observed how many recorded outcomes fell inside that set, and divided by the total number of trials.

3.11 What frequencies teach us

Observed frequencies have structural properties. These properties are not arbitrary. They come from the way sets of outcomes are counted in repeated observations.

First, frequencies are non-negative:

$$f(A) \geq 0.$$

They cannot be negative because counts cannot be negative.

Second, frequencies are at most 1:

$$f(A) \leq 1.$$

The event A cannot occur more often than the total number of trials.

Third, the whole sample space occurs on every trial:

$$f(\Omega) = 1.$$

Every recorded outcome belongs to Ω by construction.

Fourth, the impossible event never occurs:

$$f(\varnothing) = 0.$$

No recorded outcome belongs to the empty set.

The most important structural property concerns disjoint events. If $A \cap B = \varnothing$, then no trial can contribute to both A and B. Therefore

$$n(A \cup B) = n(A) + n(B),$$

and after division by the total number of trials,

$$f(A \cup B) = f(A) + f(B).$$

Theorem

For observed frequencies on a fixed finite sample space:

$$f(\varnothing) = 0, \qquad f(\Omega) = 1,$$

$$0 \leq f(A) \leq 1,$$

and if $A \cap B = \varnothing$, then

$$f(A \cup B) = f(A) + f(B).$$

These are not yet axioms. They are observations about how empirical frequencies behave. The axioms of probability will later abstract exactly this pattern.

3.12 When simple addition fails

If two events are not disjoint, simple addition counts the overlap twice. This is not a numerical accident; it is a structural fact about sets.

Let

$$M = \{1, 2, 3\}, \qquad N = \{3, 4, 5\}.$$

Then

$$M \cap N = \{3\}.$$

If we add $n(M)$ and $n(N)$, the trials with outcome 3 are counted once as part of M and once as part of N. Therefore

$$n(M \cup N) = n(M) + n(N) - n(M \cap N).$$

At the frequency level,

$$f(M \cup N) = f(M) + f(N) - f(M \cap N).$$

Remark

The correction term is not a trick. It is the price of overlap. Whenever an outcome has two reasons to be included, addition counts it twice unless the overlap is subtracted once.

The philosophy remains the same: the numerical rule is not primary. The structure of the event comes first; the numerical rule follows from that structure.

3.13 From observed frequency to probability

Observed frequencies are tied to a particular record of trials. If the same experiment is repeated again, the frequencies may change slightly. Yet the structure suggested by frequencies is stable:

events receive numbers between 0 and 1, the whole sample space receives 1, the empty event receives 0, and disjoint unions receive sums.

This suggests a new mathematical object. Instead of recording what happened in one finite collection of trials, we introduce a function that assigns a number to each event in a way that preserves the structural rules discovered above.

Definition

A **probability assignment** on a finite sample space Ω is a function

$$P : \mathcal{P}(\Omega) \to [0, 1]$$

that assigns a number $P(A)$ to each event $A \subseteq \Omega$.

The notation $\mathcal{P}(\Omega)$ denotes the power set of Ω, that is, the set of all subsets of Ω. Since events are subsets of Ω, the domain of P is the collection of events.

But not every function from $\mathcal{P}(\Omega)$ to $[0, 1]$ deserves to be called probability. The function must respect the structure of events.

Theorem

In a finite setting, a probability assignment should satisfy:

1. **Non-negativity:** for every event A,
$$P(A) \geq 0.$$
2. **Normalization:**
$$P(\Omega) = 1.$$
3. **Additivity for disjoint events:** if $A \cap B = \varnothing$, then
$$P(A \cup B) = P(A) + P(B).$$

These properties are not decorative. They are the minimum conditions needed for the numerical language to remain faithful to the set language.

3.14 Consequences of the axioms in the finite case

Once the axioms are accepted, several familiar rules become consequences rather than assumptions.

First, the impossible event has probability zero. Since

$$\Omega = \Omega \cup \varnothing$$

and the union is disjoint, additivity gives

$$P(\Omega) = P(\Omega) + P(\varnothing).$$

Subtracting $P(\Omega)$ from both sides gives

$$P(\varnothing) = 0.$$

Second, complements add up to one. For any event A, the events A and A^c are disjoint and

$$A \cup A^c = \Omega.$$

Therefore

$$P(A) + P(A^c) = P(\Omega) = 1,$$

so

$$P(A^c) = 1 - P(A).$$

Third, if $A \subseteq B$, then $P(A) \le P(B)$. Indeed, B can be decomposed as

$$B = A \cup (B \setminus A),$$

where the two parts are disjoint. Hence

$$P(B) = P(A) + P(B \setminus A).$$

Since probabilities are non-negative,

$$P(B \setminus A) \ge 0,$$

and therefore

$$P(B) \ge P(A).$$

Finally, for arbitrary events A and B, the union formula follows from splitting the union into disjoint pieces:

$$A \cup B = A \cup (B \setminus A).$$

Thus

$$P(A \cup B) = P(A) + P(B \setminus A).$$

But B itself decomposes as

$$B = (B \setminus A) \cup (A \cap B),$$

so

$$P(B) = P(B \setminus A) + P(A \cap B).$$

Therefore

$$P(B \setminus A) = P(B) - P(A \cap B),$$

and hence

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Remark

The formulas are not separate facts to memorize. They are consequences of preserving the structure of events. Once the event algebra is understood, the probability rules become almost inevitable.

3.15 The axiomatic point of view

The modern theory of probability is axiomatic. This does not mean that it ignores intuition. It means that it takes the stable part of intuition and turns it into a structure that can be used reliably.

In finite spaces, the essential ideas are already visible:

- events are subsets of a sample space;
- probability assigns numbers to events;
- the whole space has probability 1;

- disjoint alternatives add.

For general probability spaces, the same philosophy is retained, but one must be more careful. Instead of assigning probabilities to all subsets of Ω, one usually assigns probabilities to a chosen family of events $\mathcal{F}$. This family must be closed under the operations needed for logical reasoning: complements, countable unions, and therefore also countable intersections. Such a family is called a σ-algebra.

Definition

A **probability space** is a triple

$$(\Omega, \mathcal{F}, P),$$

where Ω is a sample space, $\mathcal{F}$ is a σ-algebra of events, and $P : \mathcal{F} \to [0, 1]$ is a probability measure satisfying:

1. $P(A) \geq 0$ for all $A \in \mathcal{F}$,
2. $P(\Omega) = 1$,
3. if $A_1, A_2, A_3, \ldots$ are pairwise disjoint events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

The third condition is countable additivity. In finite examples it reduces to the additivity rules already seen. In infinite settings it becomes a deep structural requirement: it says that probability remains compatible with decompositions into countably many disjoint alternatives.

This is the philosophical endpoint of the chapter. We began with informal statements about outcomes. We discovered that statements select sets. We discovered that logical relations between statements become set operations. We observed that frequencies assign numbers to events in a structured way. Finally, we abstracted the stable part of that structure into axioms.

The beauty of the formalism is that it is not imposed from outside. It is discovered by asking, step by step, what must be preserved if reasoning under uncertainty is to become mathematics.

Chapter 4

Sample Spaces and Elementary Probability

This chapter develops several canonical examples of sample spaces and elementary probability from the ground up. The emphasis is on understanding the experiment, deciding what should count as an elementary outcome, constructing the sample space, defining events, and assigning probabilities only after the mathematical model has been made explicit.

Different representations of the same probabilistic structure are used throughout the chapter: lists, tables, trees, diagrams, and symbolic notation. The goal is to show that a sample space is an abstract object produced by analysis and modelling, not a collection of observed data.

4.1 From an experiment to a sample space

The previous chapter introduced the formal language of probability: a sample space, events as subsets of the sample space, operations on events, and a probability assignment. We now use that language in concrete models. The main point of the present chapter is methodological:

a sample space is not found inside the experiment; it is constructed.

A physical or conceptual experiment may be described in ordinary language, but ordinary language does not by itself say what the elementary outcomes are. We must decide what information is recorded, what distinctions are visible in the model, and what distinctions are deliberately ignored.

Definition

A **probabilistic model** begins with a decision about what is recorded. A **sample space** Ω is the set of all elementary outcomes under that recording rule. A **probability assignment** then gives probabilities to events, that is, to subsets of Ω.

Thus the construction has the following order:

$$\text{experiment} \longrightarrow \text{recording rule} \longrightarrow \text{elementary outcome} \longrightarrow \Omega \longrightarrow \text{events} \longrightarrow P.$$

The order matters. If we skip the recording rule, then we do not really know what one elementary outcome is. If we do not know what one elementary outcome is, then we do not know what the sample space is. If we do not know the sample space, then symbols such as $A \subseteq \Omega$ and $P(A)$ have no precise meaning.

The same physical situation may have different sample spaces

Consider the instruction:

toss a coin twice.

This instruction describes a physical procedure, but it does not yet fully describe the mathematical model. Several recording rules are possible.

If we record the full ordered sequence of outcomes, then one elementary outcome is a sequence such as HT. The sample space is

$$\Omega_1 = \{HH, HT, TH, TT\}.$$

In this model $HT \neq TH$, because the first position and the second position are different pieces of information.

If instead we record only the number of heads, then one elementary outcome is a number. The sample space is

$$\Omega_2 = \{0, 1, 2\}.$$

In this model the sequences HT and TH are not separate elementary outcomes. They both correspond to the same recorded value, namely 1.

If we record only whether at least one head appeared, then the sample space is

$$\Omega_3 = \{\text{yes}, \text{no}\}.$$

This is another legitimate model, but it answers a different question.

Remark

The phrase "the experiment is the same" can be misleading. The physical action may be the same, but the mathematical sample space depends on what is recorded. Probability theory does not work directly with the physical action; it works with the mathematical description produced from that action.

Elementary outcomes are model-dependent

An elementary outcome is not necessarily the smallest physical object involved. It is the smallest complete result according to the model.

Example

Suppose a ball is drawn from an urn containing three red balls and two blue balls. If the balls are labelled

$$R_1, R_2, R_3, B_1, B_2,$$

and the label is recorded, then the sample space is

$$\Omega = \{R_1, R_2, R_3, B_1, B_2\}.$$

Here the elementary outcomes are individual labelled balls.

If only the colour is recorded, then the sample space is instead

$$\Omega' = \{R, B\}.$$

Here the elementary outcomes are colours, not labelled balls. The two sample spaces describe different levels of information.

The second model is not wrong. It may be exactly the model we need if the question is only about colour. But it must be treated honestly: changing the recording rule changes the sample space and may also change how probabilities are assigned.

The sample space is only half of the model

After Ω has been constructed, we still have to assign probabilities. The sample space tells us which elementary outcomes exist in the model. It does not automatically say how likely they are.

Example

For a single coin toss we may choose

$$\Omega = \{H, T\}.$$

If the coin is modelled as symmetric, then we assign

$$P(H) = P(T) = \frac{1}{2}.$$

If the coin is modelled as biased toward heads, then we may assign

$$P(H) = p, \qquad P(T) = 1 - p, \qquad p \neq \frac{1}{2}.$$

The sample space is the same in both models, but the probability assignment is different.

This distinction is fundamental. A list of possible outcomes is not yet a probability model. A probability model requires both:

$$\text{possible outcomes} \quad \text{and} \quad \text{probabilities of events.}$$

Uniform and non-uniform assignments

In a finite sample space

$$\Omega = \{\omega_1, \omega_2, \ldots, \omega_n\},$$

a probability assignment is determined by the probabilities of the elementary outcomes:

$$P(\{\omega_1\}) = p_1, \quad P(\{\omega_2\}) = p_2, \quad \ldots, \quad P(\{\omega_n\}) = p_n.$$

These numbers must satisfy

$$p_i \geq 0 \quad \text{for every } i,$$

and

$$p_1 + p_2 + \cdots + p_n = 1.$$

Then the probability of an event is obtained by adding the probabilities of the elementary outcomes contained in that event:

$$P(A) = \sum_{\omega_i \in A} p_i.$$

Definition

A finite probability model is **uniform** if all elementary outcomes have the same probability. If $|\Omega| = n$, this means

$$P(\{\omega\}) = \frac{1}{n}$$

for every $\omega \in \Omega$.

Only in a uniform finite model do we obtain the classical formula

$$P(A) = \frac{|A|}{|\Omega|}.$$

This formula is not the definition of probability in every finite model. It is a consequence of the special assumption that all elementary outcomes are equally likely.

Remark

A common error is to count how many named outcomes appear in a sample space and then assume that all names have equal probability. Equal naming is not the same as equal probability. Equal probability must come from the mechanism of the model or from an explicit modelling assumption.

Events are questions asked of the sample space

Once Ω is fixed, an event is a subset of Ω. Conceptually, an event answers a question about the elementary outcome.

Example

If a die is rolled and

$$\Omega = \{1, 2, 3, 4, 5, 6\},$$

then the question

is the result even?

selects the event

$$A = \{2, 4, 6\}.$$

The question

is the result at least 5?

selects the event

$$B = \{5, 6\}.$$

The compound question

is the result even and at least 5?

selects the event

$$A \cap B = \{6\}.$$

This is why the sample space must be fixed before events are discussed. The same sentence can select different sets in different spaces.

A first complete modelling pattern

When solving elementary probability problems, we shall repeatedly use the following pattern.

1. Describe the physical or conceptual experiment.
2. State what information is recorded.
3. Identify one elementary outcome.
4. Construct the sample space Ω.
5. Describe the event or events of interest as subsets of Ω.
6. Decide whether elementary outcomes are equally likely.
7. Assign probabilities to elementary outcomes or directly to events.
8. Compute the required probability and interpret it in the original situation.

This pattern may look slow. That is intentional. At the beginning, speed is less important than correctness. Most wrong solutions to elementary probability problems do not fail because of difficult arithmetic. They fail because the sample space, the elementary outcomes, or the probability assignment were never made explicit.

Remark

The goal of this chapter is not to collect examples. The goal is to learn a method: construct the model first, calculate only after the model is clear.

4.2 Coin models

Coin tossing is the simplest family of probability models, but it is not trivial. It already contains several ideas that will appear throughout the course: a sample space, a recording rule, a probability assignment, repeated experiments, product spaces, events described by conditions, and the difference between a particular sequence and a numerical summary of that sequence.

The physical object called a coin does not determine a probability model by itself. We must say what is recorded and what assumptions are made about the mechanism. A coin may be treated as symmetric, or it may be treated as biased. A single toss may be recorded, or a whole sequence of tosses may be recorded. Each of these choices changes the model or the probabilities inside the model.

One toss of a coin

Suppose first that a coin is tossed once and that we record which side lands up. The elementary outcome is one of two symbols:

$$H \quad \text{or} \quad T,$$

where H denotes heads and T denotes tails. The sample space is

$$\Omega = \{H, T\}.$$

At this point we have only described the possible outcomes. We have not yet assigned probabilities.

If the coin is modelled as symmetric, then the two elementary outcomes are given equal probability:

$$P(H) = P(T) = \frac{1}{2}.$$

This is a modelling assumption based on symmetry. It is not a consequence of the fact that the set Ω has two elements. The fact that $|\Omega| = 2$ tells us how many elementary outcomes there are. It does not by itself tell us how likely they are.

If the coin is biased, we may instead write

$$P(H) = p, \qquad P(T) = 1 - p,$$

where

$$0 \le p \le 1.$$

The same sample space $\{H, T\}$ now carries a different probability assignment. When $p = 1/2$, the model is symmetric. When $p \neq 1/2$, one side is more likely than the other.

Remark

The symbols H and T are not probabilities. They are elementary outcomes. The numbers assigned to them come from an additional assumption about the coin and the tossing mechanism.

Events in the one-toss model

In the one-toss model there are only four events:

$$\varnothing, \qquad \{H\}, \qquad \{T\}, \qquad \Omega.$$

Their probabilities are determined by the probabilities of H and T:

$$P(\varnothing) = 0, \qquad P(\Omega) = 1,$$

$$P(\{H\}) = p, \qquad P(\{T\}) = 1 - p.$$

For a symmetric coin this becomes

$$P(\{H\}) = P(\{T\}) = \frac{1}{2}.$$

Even this small example illustrates the general finite rule. If an event is a set of elementary outcomes, then its probability is obtained by adding the probabilities of the elementary outcomes inside it.

Two tosses: sequences as elementary outcomes

Now suppose the coin is tossed twice and the complete order of results is recorded. An elementary outcome is no longer a single symbol. It is an ordered sequence of length two. The sample space is

$$\Omega = \{HH, HT, TH, TT\}.$$

The outcomes HT and TH are different because the first position and the second position are different. The outcome HT means heads first and tails second. The outcome TH means tails first and heads second.

If the coin is symmetric and the two tosses are modelled as independent repetitions of the same mechanism, then every sequence has probability

$$\frac{1}{4}.$$

Thus

$$P(HH) = P(HT) = P(TH) = P(TT) = \frac{1}{4}.$$

If the coin has $P(H) = p$ and $P(T) = 1 - p$, then the probabilities of the four sequences are

$$P(HH) = p^2, \qquad P(HT) = p(1-p),$$

$$P(TH) = (1-p)p, \qquad P(TT) = (1-p)^2.$$

The middle two values are equal because multiplication of numbers is commutative:

$$p(1-p) = (1-p)p.$$

But the sequences HT and TH remain different elementary outcomes.

Example

Let $p = 0.7$. Then

$$P(HH) = 0.49, \qquad P(HT) = 0.21, \qquad P(TH) = 0.21, \qquad P(TT) = 0.09.$$

These probabilities add to one:

$$0.49 + 0.21 + 0.21 + 0.09 = 1.$$

The sample space has four outcomes, but the model is not uniform.

The event "exactly one head"

The event

$$A = \{\text{exactly one head occurs}\}$$

contains two elementary outcomes:

$$A = \{HT, TH\}.$$

Therefore, for a symmetric coin,

$$P(A) = P(HT) + P(TH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

For a biased coin with $P(H) = p$,

$$P(A) = p(1-p) + (1-p)p = 2p(1-p).$$

Notice the difference between an elementary outcome and an event. The sequence HT is one elementary outcome. The phrase "exactly one head" describes an event containing two elementary outcomes. Probability problems often ask for the probability of an event, not of a single elementary outcome.

Three tosses and the structure of a product space

For three tosses, with order recorded, the sample space is

$$\Omega = \{H, T\}^3.$$

This notation means the set of all ordered triples whose entries are H or T. Explicitly,

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

There are

$$2^3 = 8$$

elementary outcomes, because each of the three positions has two possible symbols.

For a symmetric coin, each sequence has probability

$$\frac{1}{8}.$$

For a biased coin with $P(H) = p$, the probability of a sequence is obtained by multiplying the probabilities of its symbols. For example,

$$P(HTH) = p(1-p)p = p^2(1-p),$$

and

$$P(TTH) = (1-p)(1-p)p = (1-p)^2 p.$$

The exponent records how many times a symbol appears in the sequence. A sequence with k heads and $3 - k$ tails has probability

$$p^k(1-p)^{3-k}.$$

The general model of n tosses

For n tosses, with order recorded, the sample space is

$$\Omega = \{H, T\}^n.$$

An elementary outcome is a sequence

$$\omega = (a_1, a_2, \ldots, a_n),$$

where each a_i is either H or T. There are

$$|\Omega| = 2^n$$

such sequences.

If the coin is symmetric and each toss is modelled as an independent repetition, then the model is uniform:

$$P(\omega) = \frac{1}{2^n}$$

for every $\omega \in \Omega$.

If $P(H) = p$ and $P(T) = 1 - p$, then a sequence with k heads and $n - k$ tails has probability

$$p^k(1 - p)^{n-k}.$$

This formula is not yet the probability of "exactly k heads". It is the probability of one particular sequence with k heads.

Example

For five tosses, the sequence

$$HHTTH$$

has three heads and two tails. Its probability in the biased model is

$$p^3(1 - p)^2.$$

The sequence

$$HTHTH$$

also has three heads and two tails, so it has the same probability

$$p^3(1 - p)^2.$$

But the two sequences are different elementary outcomes.

Exactly k heads in n tosses

Now consider the event

$$A_k = \{\omega \in \{H, T\}^n : \omega \text{ contains exactly } k \text{ heads}\}.$$

This event contains all sequences of length n in which exactly k positions are occupied by H, and the remaining $n - k$ positions are occupied by T.

To count these sequences, we do not choose the heads themselves as physical objects. We choose the positions at which heads occur. There are n positions, and we choose k of them to contain H. Once those positions are chosen, the remaining positions must contain T. Hence

$$|A_k| = \binom{n}{k}.$$

For a symmetric coin, the whole space is uniform, so

$$P(A_k) = \frac{|A_k|}{|\Omega|} = \frac{\binom{n}{k}}{2^n}.$$

For a biased coin, the space is generally not uniform. However, every sequence in A_k has the same probability $p^k(1-p)^{n-k}$. Since there are $\binom{n}{k}$ such sequences, additivity gives

$$P(A_k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Theorem

If a coin with $P(H) = p$ is tossed n times independently, then the probability of exactly k heads is

$$\binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \dots, n.$$

This formula has three parts:

- $\binom{n}{k}$ counts the positions of the heads;
- p^k is the contribution of the k heads;
- $(1-p)^{n-k}$ is the contribution of the $n-k$ tails.

What this model teaches

The coin model teaches several lessons that will recur in more complicated examples.

First, the sample space depends on what is recorded. One toss gives $\{H, T\}$; n ordered tosses give $\{H, T\}^n$; recording only the number of heads gives $\{0, 1, \dots, n\}$.

Second, the probability assignment is separate from the sample space. The same space $\{H, T\}$ can carry the symmetric assignment $1/2, 1/2$ or the biased assignment $p, 1-p$.

Third, an event may contain many elementary outcomes. The event "exactly k heads" is not one sequence; it is a set of sequences.

Fourth, combinatorics enters probability when we count how many elementary outcomes satisfy a condition. The binomial coefficient appears because we count positions of successes, not because it is a formula to be memorized in isolation.

Remark

The coin model is simple enough to be completely transparent, but rich enough to show the entire modelling workflow: choose the recording rule, construct Ω, assign probabilities, define events, count favourable outcomes, and interpret the result.

4.3 Dice models

Dice models are the natural next step after coin models. A die has more than two visible faces, and this makes it easier to see the difference between a uniform sample space and a non-uniform probability assignment. Dice also provide one of the most important examples in elementary probability: the model of two dice, where elementary outcomes are ordered pairs but many questions are about sums, comparisons, or other functions of those pairs.

As before, we do not begin with a formula. We begin with the experiment and the recording rule.

One roll of one die

Suppose a die is rolled once and the number on the upper face is recorded. The sample space is

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

An elementary outcome is one number from this set.

If the die is modelled as fair, then all six elementary outcomes have the same probability:

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}.$$

This is the uniform model on a six-element sample space.

If the die is not assumed to be fair, then we may write

$$P(i) = p_i, \qquad i = 1, 2, 3, 4, 5, 6,$$

where

$$p_i \geq 0$$

for every i, and

$$p_1 + p_2 + p_3 + p_4 + p_5 + p_6 = 1.$$

The sample space is still $\{1, 2, 3, 4, 5, 6\}$, but the probability assignment is no longer necessarily uniform.

Remark

The sentence "a die has six possible results" does not imply that each result has probability $1/6$. The value $1/6$ comes from the additional modelling assumption that the die and the rolling mechanism are symmetric enough to treat the six faces equally.

Events for one die

Events are subsets of Ω. For example, the event that the result is even is

$$E = \{2, 4, 6\}.$$

The event that the result is at least 5 is

$$A = \{5, 6\}.$$

The event that the result is prime is

$$B = \{2, 3, 5\}.$$

In the fair model,

$$P(E) = \frac{|E|}{|\Omega|} = \frac{3}{6} = \frac{1}{2},$$

$$P(A) = \frac{2}{6} = \frac{1}{3},$$

and

$$P(B) = \frac{3}{6} = \frac{1}{2}.$$

In a non-uniform model, we cannot use only the number of favourable outcomes. Instead, we must add the probabilities of the elementary outcomes inside the event:

$$P(E) = p_2 + p_4 + p_6,$$

$$P(A) = p_5 + p_6,$$

$$P(B) = p_2 + p_3 + p_5.$$

Example

Suppose
$$(p_1, p_2, p_3, p_4, p_5, p_6) = (0.10, 0.10, 0.15, 0.15, 0.20, 0.30).$$
Then the event $E = \{2, 4, 6\}$ has probability
$$P(E) = 0.10 + 0.15 + 0.30 = 0.55.$$
Even though E contains three of the six possible faces, its probability is not $3/6$, because the elementary outcomes are not equally likely.

Two dice as ordered pairs

Now suppose two dice are rolled and the number on each die is recorded. If the dice are distinguished, for example by colour, then the outcome is an ordered pair
$$(i, j),$$
where i is the result on the first die and j is the result on the second die. The sample space is
$$\Omega = \{1, 2, 3, 4, 5, 6\} \times \{1, 2, 3, 4, 5, 6\}.$$
Equivalently,
$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}.$$
It has
$$|\Omega| = 6 \cdot 6 = 36$$
elementary outcomes.

If both dice are fair and the rolls are modelled as independent, then every ordered pair has probability
$$\frac{1}{36}.$$
Thus the model is uniform on the set of ordered pairs.

Remark

The outcomes $(2, 5)$ and $(5, 2)$ are different in this model. They represent different results on the first and second die. They may lead to the same sum, but they are not the same elementary outcome.

The event that the sum is seven

Consider the event
$$S_7 = \{\text{the sum of the two dice is } 7\}.$$
In the ordered-pair model this event is the set
$$S_7 = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$
It contains six elementary outcomes. Since the fair two-dice model is uniform,
$$P(S_7) = \frac{|S_7|}{|\Omega|} = \frac{6}{36} = \frac{1}{6}.$$

The count 6 comes from a structural fact. To obtain sum 7, once the first coordinate i is chosen, the second coordinate must be $7 - i$. The possible first coordinates are $1, 2, 3, 4, 5, 6$, and each gives one valid ordered pair.

Example

The event that the sum is 4 is

$$S_4 = \{(1,3),(2,2),(3,1)\}.$$

Hence

$$P(S_4) = \frac{3}{36} = \frac{1}{12}$$

in the fair two-dice model.

The events S_4 and S_7 are both described by a sum, but they contain different numbers of ordered pairs. Therefore they have different probabilities in the fair two-dice model.

Representing two dice by a table

A useful representation of the two-dice sample space is a 6×6 table:

	1	2	3	4	5	6
1	$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$	$(1,5)$	$(1,6)$
2	$(2,1)$	$(2,2)$	$(2,3)$	$(2,4)$	$(2,5)$	$(2,6)$
3	$(3,1)$	$(3,2)$	$(3,3)$	$(3,4)$	$(3,5)$	$(3,6)$
4	$(4,1)$	$(4,2)$	$(4,3)$	$(4,4)$	$(4,5)$	$(4,6)$
5	$(5,1)$	$(5,2)$	$(5,3)$	$(5,4)$	$(5,5)$	$(5,6)$
6	$(6,1)$	$(6,2)$	$(6,3)$	$(6,4)$	$(6,5)$	$(6,6)$

Rows record the first die and columns record the second die. This table is not just a picture. It makes the elementary outcomes visible and allows events to be seen as regions.

For example, the event

$$A = \{(i,j) : i = j\}$$

that both dice show the same number is the diagonal of the table:

$$A = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}.$$

Thus

$$P(A) = \frac{6}{36} = \frac{1}{6}.$$

The event

$$B = \{(i,j) : i > j\}$$

that the first die shows a larger number than the second die is the region below the diagonal. It contains

$$1+2+3+4+5 = 15$$

elementary outcomes, so

$$P(B) = \frac{15}{36} = \frac{5}{12}.$$

The trap of recording only the sum

There is another possible recording rule. Instead of recording the ordered pair, we might record only the sum. Then the sample space becomes

$$\Omega' = \{2,3,4,5,6,7,8,9,10,11,12\}.$$

This is a legitimate sample space if the recorded outcome is only the sum. However, it is not a uniform sample space when the sum comes from rolling two fair dice.

The value 7 can occur in six ordered ways:

$$(1,6),(2,5),(3,4),(4,3),(5,2),(6,1).$$

The value 2 can occur in only one ordered way:

$$(1,1).$$

Therefore

$$P(7)=\frac{6}{36}, \qquad P(2)=\frac{1}{36}.$$

They are not equal.

Remark

The set $\{2,3,\ldots,12\}$ has eleven elements, but the sums of two fair dice are not equally likely. A smaller or more aggregated sample space may be useful, but its probabilities must be inherited from the underlying mechanism, not assigned uniformly without justification.

The complete distribution of the sum is

s	2	3	4	5	6	7	8	9	10	11	12
# pairs	1	2	3	4	5	6	5	4	3	2	1

Hence

$$P(S=s)=\frac{\#\{(i,j):i+j=s\}}{36}.$$

For example,

$$P(S=10)=\frac{3}{36}=\frac{1}{12}.$$

Non-uniform dice and product assignments

If one die has probabilities

$$P(1)=p_1,\ldots,P(6)=p_6$$

and another die has probabilities

$$Q(1)=q_1,\ldots,Q(6)=q_6,$$

then, under the assumption that the two rolls are independent, the probability of the ordered pair (i,j) is

$$P((i,j))=p_iq_j.$$

The event that the sum is 7 then has probability

$$P(S_7)=p_1q_6+p_2q_5+p_3q_4+p_4q_3+p_5q_2+p_6q_1.$$

This expression is longer than $6/36$, but it says exactly the same kind of thing: add the probabilities of the elementary outcomes that belong to the event.

Repeated rolls of one die

For n rolls of one die, with order recorded, the sample space is

$$\Omega=\{1,2,3,4,5,6\}^n.$$

An elementary outcome is a sequence

$$(a_1,a_2,\ldots,a_n),$$

where each $a_i \in \{1, 2, 3, 4, 5, 6\}$. The number of elementary outcomes is

$$6^n.$$

If the die is fair and the rolls are independent, then each sequence has probability

$$\frac{1}{6^n}.$$

If the die has probabilities $p_1, \dots, p_6$, then

$$P(a_1, a_2, \dots, a_n) = p_{a_1} p_{a_2} \cdots p_{a_n}.$$

Example

For three rolls of a non-uniform die, the sequence $(2, 6, 2)$ has probability

$$p_2 p_6 p_2 = p_2^2 p_6.$$

The event "exactly two sixes in three rolls" contains the sequences

$$(6, 6, x), \quad (6, x, 6), \quad (x, 6, 6),$$

where x is not 6. In the fair model its probability is

$$\binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right).$$

The binomial coefficient counts the positions occupied by the sixes.

What dice models teach

Dice models teach three important lessons.

First, a finite sample space may be uniform or non-uniform. The same set $\{1, 2, 3, 4, 5, 6\}$ can describe a fair die or a biased die.

Second, when an experiment has several stages, an elementary outcome is often a sequence or ordered tuple. For two dice, the natural elementary outcomes are ordered pairs.

Third, an aggregated value such as a sum is not automatically uniform. The sum of two fair dice has values in $\{2, 3, \dots, 12\}$, but these values have different probabilities because they correspond to different numbers of ordered pairs.

Remark

The safest modelling habit is to construct the detailed sample space first and only then decide whether a coarser description, such as the sum, is appropriate. Aggregation is useful, but it must preserve the probabilities induced by the underlying model.

4.4 Card models

Card models are useful because the phrase "draw cards" is incomplete. It does not say whether we record one card or several cards, whether order matters, whether cards are replaced, or whether the final object is a hand, a sequence, or only a summary such as the number of aces. Each of these decisions changes the sample space.

In this section let D denote a standard deck of 52 distinct cards. The cards are distinct because a card is determined by both rank and suit. For example, the ace of hearts and the ace of spades are different cards.

Drawing one card

Suppose one card is drawn from a well-shuffled deck and the exact card is recorded. The sample space is

$$\Omega = D.$$

It has

$$|\Omega| = 52$$

elementary outcomes. If the deck is well shuffled and no card is favoured, the uniform model assigns

$$P(c) = \frac{1}{52}$$

for every card $c \in D$.

Events are subsets of the deck. For example, the event that the card is an ace is

$$A = \{\text{ace of hearts}, \text{ace of diamonds}, \text{ace of clubs}, \text{ace of spades}\}.$$

Therefore

$$P(A) = \frac{4}{52} = \frac{1}{13}.$$

The event that the card is a heart contains 13 cards, so

$$P(\text{heart}) = \frac{13}{52} = \frac{1}{4}.$$

The event that the card is a face card contains the jacks, queens, and kings in four suits, hence 12 cards, so

$$P(\text{face card}) = \frac{12}{52} = \frac{3}{13}.$$

Remark

The sample space contains individual cards, not ranks. If we record the exact card, then there are 52 elementary outcomes. The event "ace" is not one elementary outcome; it is a set of four elementary outcomes.

A five-card hand as a set

Now suppose five cards are dealt and only the final hand is recorded. The order in which the cards arrived is ignored. Then an elementary outcome is a five-card subset of the deck:

$$S \subseteq D, \qquad |S| = 5.$$

The sample space is

$$\Omega = \{S \subseteq D : |S| = 5\}.$$

Its size is

$$|\Omega| = \binom{52}{5}.$$

This count appears because a hand is a set. We choose which five cards belong to the hand, and the order of listing those cards does not create a new hand. Thus

$$\{A\heartsuit, K\clubsuit, 7\diamondsuit, 3\spadesuit, 2\heartsuit\}$$

is the same hand no matter in which order we write its elements.

If the deck is well shuffled and every five-card hand is equally likely, then the uniform formula applies:

$$P(E) = \frac{|E|}{\binom{52}{5}}$$

for any event E consisting of five-card hands.

Example

Let E be the event that the hand contains exactly one ace. To construct such a hand, choose one of the four aces and choose the remaining four cards from the 48 non-aces. Hence

$$|E| = \binom{4}{1}\binom{48}{4}.$$

Therefore

$$P(E) = \frac{\binom{4}{1}\binom{48}{4}}{\binom{52}{5}}.$$

This is a probability of an event, not of a single hand. The event contains many hands.

Five cards drawn in order

Suppose instead that five cards are drawn one after another and the order is recorded. Then an elementary outcome is a sequence

$$(c_1, c_2, c_3, c_4, c_5),$$

where all five cards are distinct. The sample space is

$$\Omega_{\text{seq}} = \{(c_1, c_2, c_3, c_4, c_5) : c_i \in D,\ c_i \neq c_j \text{ for } i \neq j\}.$$

Its size is

$$52 \cdot 51 \cdot 50 \cdot 49 \cdot 48.$$

The first position has 52 choices, the second has 51 choices because one card has already been used, and so on.

This sample space is different from the five-card hand space. A hand ignores order; a sequence records order. The same five cards can appear in

$$5!$$

different orders. Therefore

$$52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 = 5!\binom{52}{5}.$$

Remark

The phrase "draw five cards" is not enough to determine the sample space. If order is ignored, the elementary outcome is a set. If order is recorded, the elementary outcome is a sequence. Both models are legitimate, but they answer different questions.

The same event in two sample spaces

Consider the statement

the five cards contain exactly one ace.

In the hand model, the event is a set of five-card subsets. In the sequence model, the event is a set of ordered five-tuples. The verbal statement is the same, but the formal event lives in a different sample space.

In the hand model we already counted

$$\binom{4}{1}\binom{48}{4}$$

favourable hands.

In the ordered model, we may count favourable sequences directly. First choose which position contains the ace: 5 choices. Choose the ace: 4 choices. Then fill the remaining four ordered positions with non-aces without repetition:

$$48 \cdot 47 \cdot 46 \cdot 45$$

choices. Thus the number of favourable sequences is

$$5 \cdot 4 \cdot 48 \cdot 47 \cdot 46 \cdot 45.$$

The total number of ordered sequences is

$$52 \cdot 51 \cdot 50 \cdot 49 \cdot 48.$$

So the probability in the ordered model is

$$\frac{5 \cdot 4 \cdot 48 \cdot 47 \cdot 46 \cdot 45}{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}.$$

This value is equal to

$$\frac{\binom{4}{1}\binom{48}{4}}{\binom{52}{5}},$$

because both models describe the same random dealing mechanism at different levels of detail.

The equality is not magic. Each unordered hand corresponds to exactly $5!$ ordered sequences. The factor $5!$ appears in both the favourable count and the total count, so it cancels.

Drawing with replacement

A different model appears if each drawn card is returned to the deck before the next draw. Suppose five draws are made, the exact card is recorded each time, and replacement occurs after every draw. Then the same card may appear several times in the recorded sequence. The sample space is

$$\Omega_{\text{rep}} = D^5.$$

Its size is

$$|\Omega_{\text{rep}}| = 52^5.$$

If the deck is well shuffled before each draw, every sequence in D^5 has probability

$$\frac{1}{52^5}.$$

This is not the same as drawing five cards without replacement. For example, under replacement the sequence

$$(A\heartsuit, A\heartsuit, 7\clubsuit, 2\diamondsuit, K\spadesuit)$$

is possible. Without replacement it is impossible, because the same physical card cannot be drawn twice.

Example

In five draws with replacement, what is the probability that all five cards are aces? Each draw has probability $4/52 = 1/13$ of producing an ace. Since replacement makes the same probability available at each draw, the probability is

$$\left(\frac{4}{52}\right)^5 = \left(\frac{1}{13}\right)^5.$$

Equivalently, there are 4^5 ace-only sequences and 52^5 total sequences, so

$$P(\text{all aces}) = \frac{4^5}{52^5}.$$

Without replacement, the probability that five drawn cards are all aces is zero, because the deck contains only four aces. The change in the physical procedure changes the sample space and therefore changes the probabilities.

At least one ace

Card models are also useful for showing the complement method. In a five-card hand, let

$$A = \{\text{the hand contains at least one ace}\}.$$

Counting hands with at least one ace directly would require cases: exactly one ace, exactly two aces, exactly three aces, exactly four aces. It is often simpler to count the complement.

The complement is

$$A^c = \{\text{the hand contains no aces}\}.$$

There are 48 non-aces, so

$$|A^c| = \binom{48}{5}.$$

Therefore

$$P(A^c) = \frac{\binom{48}{5}}{\binom{52}{5}},$$

and

$$P(A) = 1 - P(A^c) = 1 - \frac{\binom{48}{5}}{\binom{52}{5}}.$$

The complement method is not a trick. It is event algebra: instead of counting all hands that satisfy the event, we count the hands that fail it and subtract from the whole space.

What card models teach

Card models emphasize that the same everyday phrase can hide several different mathematical structures.

If one card is drawn and the exact card is recorded, then $\Omega = D$. If five cards are recorded as a hand, then Ω is a set of five-element subsets. If five cards are recorded in order without replacement, then Ω is a set of sequences with no repeated card. If cards are drawn with replacement, then $\Omega = D^5$, and repeated cards become possible.

Remark

Before solving a card problem, always identify whether the elementary outcome is a single card, a set of cards, an ordered sequence of cards, or a summary of the cards. Most card-counting errors are errors about the sample space, not errors about arithmetic.

4.5 Urn models

Urn models are a standard way to study how probability spaces are constructed. They are especially useful because one physical setup can be described at several levels. We may record the exact labelled object selected from the urn, or we may record only a visible type such as colour. We may sample with replacement or without replacement. We may record order or ignore order. Each choice leads to a different sample space.

One selection from labelled objects

Suppose an urn contains five labelled objects:

$$R_1, R_2, R_3, B_1, B_2,$$

where the letter records type and the subscript records the label. If one object is selected and its label is recorded, then

$$\Omega = \{R_1, R_2, R_3, B_1, B_2\}.$$

If the selection mechanism treats the five labelled objects equally, then each has probability 1/5.

The event that the selected object is of type R is

$$A = \{R_1, R_2, R_3\}.$$

Therefore

$$P(A) = \frac{3}{5}.$$

The event that the selected object is of type B is

$$A^c = \{B_1, B_2\},$$

so

$$P(A^c) = \frac{2}{5}.$$

Remark

In this model, R_1, R_2, and R_3 are different elementary outcomes. They have the same type, but the labels are recorded. The event "type R" is therefore a set of elementary outcomes, not a single elementary outcome.

Recording only type

Now use the same physical urn, but record only the type of the selected object. Then the sample space is

$$\Omega' = \{R, B\}.$$

This sample space has two elements, but it is not automatically uniform. The probability of type R is inherited from the three labelled objects of type R, and the probability of type B is inherited from the two labelled objects of type B:

$$P(R) = \frac{3}{5}, \qquad P(B) = \frac{2}{5}.$$

This is the central lesson of the urn model: when several detailed elementary outcomes are grouped into one recorded outcome, equal probabilities at the detailed level may become unequal probabilities at the aggregated level.

Example

Writing

$$P(R) = P(B) = \frac{1}{2}$$

would describe a different mechanism: one that first chooses a type uniformly. It would not describe uniform selection from an urn containing three labelled objects of type R and two labelled objects of type B.

Two selections with replacement

Suppose one object is selected, recorded, returned to the urn, and then another selection is made. If labels and order are recorded, then one elementary outcome is an ordered pair

$$(x_1, x_2),$$

where each coordinate belongs to

$$\{R_1, R_2, R_3, B_1, B_2\}.$$

Because the first selected object is returned, the same label may appear twice. Thus

$$\Omega = \{R_1, R_2, R_3, B_1, B_2\}^2, \qquad |\Omega| = 25.$$

The event that both selected objects have type R is

$$A = \{(R_i, R_j) : i, j \in \{1, 2, 3\}\}.$$

There are $3 \cdot 3 = 9$ such ordered pairs, so

$$P(A) = \frac{9}{25}.$$

Equivalently,

$$P(A) = \frac{3}{5} \cdot \frac{3}{5}.$$

Two selections without replacement

Now suppose the first selected object is not returned before the second selection. If labels and order are recorded, then

$$\Omega = \{(x_1, x_2) : x_1, x_2 \in \{R_1, R_2, R_3, B_1, B_2\},\ x_1 \neq x_2\}.$$

There are

$$5 \cdot 4 = 20$$

elementary outcomes.

The event that both selected objects have type R is

$$A = \{(R_i, R_j) : i, j \in \{1, 2, 3\},\ i \neq j\}.$$

There are $3 \cdot 2 = 6$ such ordered pairs, hence

$$P(A) = \frac{6}{20} = \frac{3}{10}.$$

The probability is smaller than in the replacement model because, after one object of type R has been selected, fewer such objects remain available.

Example

The event "one object of each type" has two ordered forms:

$$(R_i, B_j) \quad \text{or} \quad (B_j, R_i).$$

There are $3 \cdot 2 = 6$ outcomes of the first form and $2 \cdot 3 = 6$ outcomes of the second form. Therefore

$$P(\text{one of each type}) = \frac{12}{20} = \frac{3}{5}.$$

Order recorded or ignored

If two objects are selected without replacement and order is ignored, then an elementary outcome is a two-element subset of the five labelled objects:

$$\Omega = \{S \subseteq \{R_1, R_2, R_3, B_1, B_2\} : |S| = 2\}.$$

Thus

$$|\Omega| = \binom{5}{2} = 10.$$

The event that both selected objects have type R has

$$\binom{3}{2} = 3$$

favourable subsets, so

$$P(\text{both type } R) = \frac{\binom{3}{2}}{\binom{5}{2}} = \frac{3}{10}.$$

This agrees with the ordered calculation because each unordered pair corresponds to exactly two ordered pairs.

Only the number of type R objects is recorded

Sometimes we ignore labels and order and record only a count. In two selections with replacement, the possible recorded values for the number of type R objects are

$$\Omega' = \{0, 1, 2\}.$$

The probabilities are not uniform. Since $P(R) = 3/5$ and $P(B) = 2/5$,

$$P(0) = \left(\frac{2}{5}\right)^2 = \frac{4}{25},$$

$$P(1) = 2 \cdot \frac{3}{5} \cdot \frac{2}{5} = \frac{12}{25},$$

$$P(2) = \left(\frac{3}{5}\right)^2 = \frac{9}{25}.$$

The middle value has a factor 2 because exactly one type R object can appear in two orders: first or second.

General formula with replacement

Suppose the urn contains r labelled objects of type R and b labelled objects of type B. If one object is selected uniformly and only type is recorded, then

$$P(R) = \frac{r}{r+b}, \qquad P(B) = \frac{b}{r+b}.$$

If n selections are made with replacement and types are recorded in order, then

$$\Omega = \{R, B\}^n.$$

The probability of exactly k recorded symbols of type R is

$$\binom{n}{k} \left(\frac{r}{r+b}\right)^k \left(\frac{b}{r+b}\right)^{n-k}.$$

The binomial coefficient counts the positions occupied by type R.

General formula without replacement

If n objects are selected without replacement and order is ignored, then the total number of possible labelled selections is

$$\binom{r+b}{n}.$$

To obtain exactly k objects of type R, choose k from the r objects of type R, and choose $n-k$ from the b objects of type B. Thus

$$P(\text{exactly } k \text{ of type } R) = \frac{\binom{r}{k}\binom{b}{n-k}}{\binom{r+b}{n}}.$$

Example

If $r = 7$, $b = 5$, and $n = 4$, then the probability of selecting exactly two objects of type R, without replacement and ignoring order, is

$$\frac{\binom{7}{2}\binom{5}{2}}{\binom{12}{4}}.$$

What urn models teach

Urn models show that probability depends on both the mechanism and the recorded information. A detailed uniform model may become non-uniform after aggregation. Replacement keeps the composition of the urn fixed from selection to selection; without replacement, the composition changes. Recording order gives sequences; ignoring order gives sets or count summaries.

Remark

The phrase "select from an urn" is not yet a probability model. A complete model must say what is recorded, whether replacement occurs, whether order is recorded, and whether labelled objects or only their types are treated as elementary outcomes.

4.6 Waiting for the first success

All examples so far have used finite sample spaces. We now return to the coin model, but change the experiment. Instead of deciding in advance how many tosses will be made, we toss until a certain event happens. This produces a sample space that is infinite but still discrete.

This example is important because it shows that a sample space does not have to be finite. It also shows why a probability model must describe not only the possible outcomes, but also the rule that generates them.

The experiment

Suppose a coin is tossed repeatedly until the first head appears. We stop as soon as the first head occurs. Assume

$$P(H) = p, \qquad P(T) = 1 - p, \qquad 0 < p \leq 1.$$

Here heads will be interpreted as success and tails as failure.

The physical experiment is not a fixed number of tosses. It may stop after one toss, or after two tosses, or after three tosses, and so on. There is no largest possible stopping time.

Recording the whole sequence

One natural recording rule is to record the whole sequence observed up to and including the first head. Then the possible elementary outcomes are

$$H, \quad TH, \quad TTH, \quad TTTH, \quad \ldots$$

The sample space is

$$\Omega = \{H, TH, TTH, TTTH, \ldots\}.$$

Equivalently,

$$\Omega = \{T^{k-1}H : k = 1, 2, 3, \ldots\},$$

where $T^{k-1}H$ means $k - 1$ tails followed by one head.

For example:

$$k = 1 \quad \Rightarrow \quad H,$$
$$k = 2 \quad \Rightarrow \quad TH,$$
$$k = 5 \quad \Rightarrow \quad TTTTH.$$

This sample space is countably infinite. Its elements can be listed in a sequence, but the list never ends.

Recording the stopping time

Another natural recording rule is to record only the number of the toss on which the first head appears. Let

$$N = \text{the number of tosses needed to obtain the first head.}$$

Then the sample space is

$$\Omega' = \{1, 2, 3, \ldots\}.$$

The outcome $N = 1$ corresponds to the sequence H. The outcome $N = 2$ corresponds to TH. The outcome $N = k$ corresponds to

$$\underbrace{TT\cdots T}_{k-1 \text{ tails}}H.$$

The two descriptions are equivalent for this experiment. Recording the full stopped sequence and recording the stopping time contain the same information, because once the stopping time is known, the sequence must consist of failures before the last toss and a success at the last toss.

Remark

The sample space $\{1, 2, 3, \ldots\}$ is not the set of all possible numbers in the world. It is the set of possible stopping times for this particular experiment.

Deriving the probability formula

We now derive the probability that the first head appears on toss k. The condition

$$N = k$$

means that the first $k-1$ tosses are tails and the k-th toss is heads. Thus the corresponding sequence is

$$\underbrace{TT\cdots T}_{k-1 \text{ times}}H.$$

The probability of one tail is $1-p$, and the probability of one head is p. Therefore

$$P(N=k) = \underbrace{(1-p)(1-p)\cdots(1-p)}_{k-1 \text{ factors}}\,p.$$

Hence

$$P(N=k) = (1-p)^{k-1}p, \qquad k = 1, 2, 3, \ldots.$$

Theorem

If independent tosses are repeated until the first success, and each toss has success probability p, then the probability that the first success occurs on trial k is

$$P(N=k) = (1-p)^{k-1}p, \qquad k = 1, 2, 3, \ldots.$$

This is called the geometric model. In this chapter we do not need the name as much as the construction. The formula appears because the outcome $N = k$ requires exactly $k-1$ failures followed by one success.

Examples

Example

Suppose the coin is fair, so $p = 1/2$. Then

$$P(N=1) = \frac{1}{2},$$

$$P(N=2) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4},$$
$$P(N=3) = \frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{8},$$

and in general

$$P(N=k) = \left(\frac{1}{2}\right)^k.$$

For a fair coin, the probability halves each time we delay the first head by one more toss.

Example

Suppose $P(H) = 0.3$. Then $P(T) = 0.7$, and

$$P(N = 4) = 0.7^3 \cdot 0.3.$$

This is the probability of the sequence

$$TTTH.$$

The first three tosses must fail, and the fourth toss must succeed.

Checking that the probabilities add to one

A probability assignment on a countable sample space should assign total probability one to the whole space. Here the elementary probabilities are

$$p, \quad (1-p)p, \quad (1-p)^2 p, \quad (1-p)^3 p, \quad \dots$$

Their sum is

$$p + (1-p)p + (1-p)^2 p + (1-p)^3 p + \cdots .$$

Factor out p:

$$p\left[1 + (1-p) + (1-p)^2 + (1-p)^3 + \cdots\right].$$

The expression in brackets is a geometric series with ratio $1-p$. Since $0 < p \leq 1$, we have $0 \leq 1-p < 1$, and

$$1 + (1-p) + (1-p)^2 + \cdots = \frac{1}{1-(1-p)} = \frac{1}{p}.$$

Therefore

$$p \cdot \frac{1}{p} = 1.$$

Thus the probabilities of all possible stopping times add to one.

Remark

This calculation also explains why we require $p > 0$. If $p = 0$, success never occurs, and the experiment "wait until the first success" does not produce a finite stopping time.

Events involving the stopping time

Once the stopping time model is built, events are subsets of

$$\{1, 2, 3, \dots\}.$$

For example, the event that the first success occurs no later than the third trial is

$$\{N \leq 3\} = \{1, 2, 3\}.$$

Its probability is

$$P(N \leq 3) = p + (1-p)p + (1-p)^2 p.$$

There is also a faster way to compute it. The complement of $\{N \leq 3\}$ is

$$\{N > 3\},$$

which means that the first three trials are all failures. Hence

$$P(N > 3) = (1 - p)^3.$$

Therefore

$$P(N \leq 3) = 1 - (1 - p)^3.$$

More generally,

$$P(N \leq m) = 1 - (1 - p)^m.$$

This is the probability that at least one success occurs in the first m trials.

Similarly,

$$P(N > m) = (1 - p)^m,$$

because the first m trials must all be failures.

Waiting for the r-th success

A natural extension is to wait not for the first success, but for the r-th success. Let

$$N_r = \text{the trial on which the } r\text{-th success occurs.}$$

For $N_r = n$ to happen, the n-th trial must be a success, and among the first $n - 1$ trials there must be exactly $r - 1$ successes.

The first $n - 1$ trials may contain those $r - 1$ successes in

$$\binom{n-1}{r-1}$$

ways. Each full sequence with r successes and $n - r$ failures has probability

$$p^r(1 - p)^{n-r}.$$

Therefore

$$P(N_r = n) = \binom{n-1}{r-1} p^r(1 - p)^{n-r}, \qquad n = r, r + 1, r + 2, \ldots.$$

Example

Suppose we wait for the second head. The event $N_2 = 5$ means that the fifth toss is heads and exactly one of the first four tosses is heads. The position of that earlier head can be chosen in

$$\binom{4}{1} = 4$$

ways. Hence

$$P(N_2 = 5) = \binom{4}{1} p^2(1 - p)^3.$$

This extension is included to show that the first-success model is not an isolated trick. It is the first member of a family of waiting-time models. The logic is always the same: describe what must happen before the stopping trial, describe what must happen at the stopping trial, count the possible positions of earlier successes, and multiply the appropriate probabilities.

What waiting-time models teach

The waiting-time model teaches several new lessons.

First, a sample space may be infinite. The set $\{1, 2, 3, \ldots\}$ is a valid discrete sample space when it records a stopping time.

Second, an infinite sample space still requires probabilities that add to one. In the first-success model, this is guaranteed by the geometric series.

Third, the elementary outcome may be represented in different but equivalent ways. We may record the stopped sequence $T^{k-1}H$, or we may record the number k. The choice depends on which description is more useful.

Fourth, a formula such as

$$P(N = k) = (1 - p)^{k-1}p$$

should be read as a description of the required pattern: failures before the stopping trial and success at the stopping trial.

Remark

Waiting-time models are a bridge between elementary finite probability and later probability distributions. They show that probability spaces can be built from a process that stops at a random time, and that even an infinite model can be handled by careful construction from elementary outcomes.

4.7 A first continuous model: uniform distribution on an interval

All models constructed so far in this chapter were discrete. Some of them were finite, such as dice, cards, and urns. One of them was countably infinite: the waiting time until the first success. These examples are essential, but they do not yet represent the full scope of probability. Many random quantities are naturally modelled as points on a continuum.

The purpose of this section is to introduce one continuous model carefully: the uniform distribution on an interval. We shall not develop the full measure theory behind continuous probability. That would require a separate course. However, we must already see the basic change in viewpoint:

in a continuous model, individual points are not the right objects to count.

Instead of counting elementary outcomes, we measure sizes of sets. For an interval on the real line, the relevant size is length.

Choosing a point from an interval

Suppose an experiment produces a real number between 0 and 1. For example, imagine an idealized pointer that lands somewhere on a line segment of unit length. The sample space is

$$\Omega = [0, 1].$$

An elementary outcome is a single real number

$$x \in [0, 1].$$

This sample space is not finite and not countably infinite. It is uncountable. There is no list

$$x_1, x_2, x_3, \ldots$$

that contains all real numbers in $[0, 1]$. This already makes the model qualitatively different from coin tosses, dice rolls, card hands, urn selections, and waiting times.

Remark

The phrase "choose a point uniformly from $[0,1]$" cannot mean that every point receives the same positive probability. There are too many points. The probability is spread continuously over the interval.

Why points have probability zero

In a finite uniform model, each elementary outcome has probability

$$\frac{1}{|\Omega|}.$$

This idea cannot be copied directly to $[0,1]$, because $[0,1]$ has uncountably many points. If each point had the same positive probability, then the total probability would be far larger than 1. If each point had probability 0, then adding point probabilities one by one cannot recover the probability of an interval.

In the continuous uniform model, every individual point has probability zero:

$$P(\{x\}) = 0$$

for each $x \in [0,1]$. This does not mean that the experiment cannot produce a point. It means that probability is not assigned by counting individual points. The probability of landing in a region is determined by the length of that region.

For example, the event

$$A = [0.2, 0.5]$$

has length 0.3, so in the uniform model on $[0,1]$ we assign

$$P(A) = 0.3.$$

The event contains infinitely many points, but its probability is controlled by how much of the interval it occupies.

Intervals as events

The simplest events in a continuous interval model are intervals. In the uniform model on $[0,1]$, the probability of an interval is its length. Thus

$$P([c,d]) = d - c$$

whenever

$$0 \le c \le d \le 1.$$

The same value is assigned to

$$(c,d), \qquad [c,d), \qquad (c,d].$$

The endpoints do not change the probability, because individual points have probability zero.

Example

Let X be uniformly distributed on $[0,1]$. Then

$$P(0.25 \le X \le 0.75) = 0.75 - 0.25 = 0.5.$$

Also,

$$P(0.25 < X < 0.75) = 0.5.$$

The difference between the closed and open interval consists only of endpoints, and endpoints have probability zero in this continuous model.

This is a major difference from discrete probability. In a discrete model, removing one elementary outcome may change the probability. In a continuous uniform model, removing finitely many points from an interval does not change its probability.

Uniform distribution on $[a, b]$

The same idea works on any finite interval

$$[a, b], \qquad a < b.$$

The sample space is

$$\Omega = [a, b].$$

The total length of the sample space is

$$b - a.$$

If the model is uniform, then probability is proportional to length. For an interval $[c, d] \subseteq [a, b]$, we define

$$P([c, d]) = \frac{d - c}{b - a}.$$

The denominator is the length of the whole sample space. The numerator is the length of the event. Thus the formula is the continuous analogue of the uniform finite formula

$$P(A) = \frac{|A|}{|\Omega|}.$$

The symbol $|A|$ counted outcomes in a finite space. In the interval model, length replaces counting.

Definition

A random point X has the **uniform distribution on** $[a, b]$ if probabilities of intervals are proportional to their lengths:

$$P(c \leq X \leq d) = \frac{d - c}{b - a}$$

for every $[c, d] \subseteq [a, b]$.

Examples

Let X be uniformly distributed on $[2, 8]$. The whole interval has length

$$8 - 2 = 6.$$

The probability that X lies between 3 and 5 is

$$P(3 \leq X \leq 5) = \frac{5 - 3}{8 - 2} = \frac{2}{6} = \frac{1}{3}.$$

The probability that X is at least 6 is

$$P(X \geq 6) = P(6 \leq X \leq 8) = \frac{8 - 6}{8 - 2} = \frac{1}{3}.$$

The probability that X is exactly 4 is

$$P(X = 4) = 0.$$

This last statement is not a contradiction. The experiment produces some exact value, but no single prescribed value occupies positive length.

Unions of intervals

Events need not be single intervals. Suppose X is uniform on $[0, 1]$, and consider the event

$$A = [0, 0.2] \cup [0.7, 1].$$

The two intervals are disjoint. Their lengths are 0.2 and 0.3. Hence

$$P(A) = 0.2 + 0.3 = 0.5.$$

This is the same additivity principle used in discrete probability: disjoint parts have probabilities that add. The difference is that the probabilities are computed from lengths rather than from counts of elementary outcomes.

What are the events in a continuous model?

For a finite or countable sample space, it is often harmless at this level to think of every subset of Ω as an event. For continuous sample spaces, this becomes too optimistic. The interval $[0, 1]$ has extremely complicated subsets, and not every subset can be assigned a length in a way that is compatible with the usual rules of probability.

Therefore, a continuous probability model is not just

$$\Omega = [0, 1].$$

It also requires a specified collection of admissible events. For the interval model, the standard choice is the collection of Borel sets, denoted informally by

$$\mathcal{B}([0, 1]).$$

These are the sets obtained from intervals by the operations that probability needs: complements, countable unions, and countable intersections.

Definition

In a continuous model, the event space is usually not taken to be the set of all subsets of Ω. Instead, one specifies a suitable collection of events, usually a **sigma-algebra**. On the real line, the standard event collection is the Borel sigma-algebra generated by open intervals.

For this course, the technical construction is less important than the modelling message. Intervals, finite unions of intervals, countable unions of intervals, and sets obtained from them by complements are legitimate events. These are more than enough for the elementary continuous examples we need now.

Remark

The word “Borel” is a signal that continuous probability requires more care than discrete probability. We introduce it here so that the model is honest: continuous probability spaces need a sample space, an event collection, and a probability assignment.

The structure of the continuous uniform model

The continuous uniform model on $[a, b]$ has three pieces:

$$\Omega = [a, b],$$

$$\mathcal{F} = \mathcal{B}([a, b]),$$

and

$$P(A) = \frac{\text{length of } A}{b - a}$$

for events A whose length is defined in the standard sense.

At this stage, we mainly use this rule for intervals and simple unions of intervals. The notation $\mathcal{F}$ reminds us that events form a selected collection of subsets of Ω, not necessarily all subsets.

What this continuous model teaches

The uniform interval model teaches several lessons that are invisible in purely discrete examples.

First, a sample space may be uncountable. The interval $[0, 1]$ contains more points than any sequence can list.

Second, probability need not be built by assigning positive mass to individual elementary outcomes. In the uniform continuous model,

$$P(\{x\}) = 0$$

for every point x, while intervals can have positive probability.

Third, the probability of an event is measured by geometric size, not by the number of points. For the uniform distribution on $[a, b]$, interval probabilities are relative lengths.

Fourth, the event collection matters. In continuous probability, we must be more careful about which subsets of the sample space are allowed as events. The standard answer on the real line is to use Borel sets.

Theorem

The continuous uniform model replaces counting by length:

$$\text{finite uniform model: } P(A) = \frac{|A|}{|\Omega|},$$

$$\text{continuous uniform interval model: } P(A) = \frac{\text{length of } A}{\text{length of } \Omega}.$$

The idea is analogous, but the mathematics is different because the sample space is uncountable.

4.8 Checklist for building a probability model

The examples in this chapter have different physical forms: coins, dice, cards, urns, and waiting times. The same modelling discipline appears in all of them. We do not begin with a formula. We begin by constructing the probability space.

The purpose of this final checklist is to make that construction explicit. When facing a new probability problem, the following questions should be answered before any numerical calculation is attempted.

Step 1: What is the experiment?

Start with the physical or conceptual procedure.

Examples include:

- toss a coin once;
- toss a coin n times;
- roll two dice;
- draw five cards from a deck;
- select objects from an urn;
- repeat trials until the first success occurs.

At this stage the description may still be incomplete. The phrase "draw five cards" does not say whether order is recorded. The phrase "select from an urn" does not say whether replacement occurs. The phrase "repeat trials" does not say what is recorded when the process stops.

Step 2: What is recorded?

The recording rule determines the mathematical outcome.

Ask:

- Is order recorded?
- Are labels recorded, or only types such as colour or rank?
- Is the whole sequence recorded, or only a numerical summary?
- Is the experiment performed with replacement or without replacement?
- Does the experiment stop after a fixed number of trials, or at a random time?

This step is essential because the same physical procedure can produce different sample spaces under different recording rules.

Remark

Most elementary probability errors begin when the recording rule is left implicit. If it is not clear what is recorded, then it is not clear what one elementary outcome is.

Step 3: What is one elementary outcome?

Once the recording rule is fixed, describe one complete elementary outcome.

Typical forms include:

- a single symbol, such as H or T;
- a number, such as a die result $i \in \{1, \dots, 6\}$;
- an ordered pair, such as (i, j) for two dice;
- a sequence, such as $HTTH$;
- a set, such as a five-card hand;
- a count, such as the number of heads;

- a stopping time, such as the trial number of the first success.

The elementary outcome must be complete relative to the recording rule. If the rule records the full sequence of coin tosses, then the number of heads is not a complete elementary outcome. If the rule records only the number of heads, then the full sequence contains more information than the model keeps.

Step 4: What is the sample space Ω?

The sample space is the set of all elementary outcomes:

$$\Omega = \{\text{all possible elementary outcomes under the chosen recording rule}\}.$$

Examples:

$$\{H, T\}, \qquad \{H, T\}^n, \qquad \{1, 2, 3, 4, 5, 6\},$$
$$\{1, \ldots, 6\}^2, \qquad \{S \subseteq D : |S| = 5\},$$
$$\{1, 2, 3, \ldots\}.$$

The sample space may be finite, such as $\{H, T\}^n$, or countably infinite, such as the waiting-time space $\{1, 2, 3, \ldots\}$.

Step 5: What events are being considered?

An event is a subset of Ω. Translate each verbal statement into a set of elementary outcomes.

Examples:

$$\text{exactly } k \text{ heads}$$

becomes

$$\{\omega \in \{H, T\}^n : \omega \text{ contains exactly } k \text{ heads}\}.$$

The statement

$$\text{the sum of two dice is 7}$$

becomes

$$\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

The statement

$$\text{a five-card hand contains at least one ace}$$

becomes a set of five-card subsets of the deck.

> **Remark**
>
> Do not count before the event has been written clearly. Counting favourable outcomes requires knowing exactly which outcomes are favourable.

Step 6: Are elementary outcomes equally likely?

This is the question that determines whether the classical formula applies.

If Ω is finite and all elementary outcomes are equally likely, then

$$P(A) = \frac{|A|}{|\Omega|}.$$

This applies, for example, to:

- a fair coin tossed n times, with full sequence recorded;

- two fair dice, with ordered pair recorded;
- a well-shuffled five-card hand, with the hand recorded as a set.

If elementary outcomes are not equally likely, then the formula

$$P(A) = \frac{|A|}{|\Omega|}$$

must not be used. Instead, add the probabilities of the elementary outcomes in A:

$$P(A) = \sum_{\omega \in A} P(\omega).$$

Examples of non-uniform models include:

- a biased coin;
- a loaded die;
- sums of two fair dice when the sample space is $\{2, 3, \ldots, 12\}$;
- colours recorded from an urn with unequal numbers of different types;
- waiting time until first success.

Step 7: Count or sum carefully

In a uniform finite model, the main task is often to count:

$$|A| \quad \text{and} \quad |\Omega|.$$

The counting method must match the structure of the sample space.

Use sequences when order matters. Use combinations when only membership matters. Use product spaces for repeated independent stages. Use complement events when the complement is easier to count.

In a non-uniform model, the main task is not only to count outcomes, but to assign and sum their probabilities. For example, for a biased coin,

$$P(\text{exactly } k \text{ heads in } n \text{ tosses}) = \binom{n}{k} p^k (1-p)^{n-k}.$$

The binomial coefficient counts the sequences, while the factor $p^k(1-p)^{n-k}$ gives the probability of each such sequence.

Step 8: Interpret the answer

A final numerical answer is not the end of the solution. It should be connected back to the original model.

For example,

$$\frac{6}{36} = \frac{1}{6}$$

for the event "sum seven on two fair dice" means that six of the thirty-six ordered pairs have sum seven. It does not mean that the value seven is one of six equally likely sums.

Similarly,

$$\frac{3}{5}$$

for drawing type R from an urn with three objects of type R and two of type B means that three of the five equally selectable labelled objects produce the recorded type R. It does not mean that the colour sample space is uniform.

A compact modelling checklist

Before calculating, write down the following:

1. What is the experiment?
2. What exactly is recorded?
3. What is one elementary outcome?
4. What is the sample space Ω?
5. What event $A \subseteq \Omega$ is being studied?
6. Are elementary outcomes equally likely?
7. If yes, what are $|A|$ and $|\Omega|$?
8. If no, what are the elementary probabilities $P(\omega)$?
9. How is $P(A)$ computed from those outcomes?
10. What does the answer mean in the original situation?

Theorem

The basic rule of this chapter is:

construct the probability model first, calculate second.

A correct probability calculation is a calculation inside a clearly described sample space with a clearly described probability assignment.

The examples in this chapter should be read as models of reasoning, not as a list of formulas. The formulas are useful because they preserve the structure of the model. When the model changes, the formula may change as well.

Chapter 5

Conditional Probability, Independence, and Bayes' Theorem

This chapter explores the algebra of events and the concept of conditional probability, which are fundamental to understanding how probabilities are updated with new information.

5.1 Information changes the space of reasoning

The previous chapter taught us how to construct probability models from experiments. We learned to identify elementary outcomes, build the sample space Ω, describe events as subsets of Ω, and assign probabilities only after the model has been made explicit.

In this chapter we add a new idea: sometimes probability is not computed from the original position of ignorance. Sometimes new information is given. When that happens, the question is no longer asked inside the whole sample space. The information restricts the set of possibilities that remain compatible with what we know.

The central message is:

new information changes the space in which we reason.

This idea is the conceptual beginning of conditional probability. We shall not begin by memorizing a formula. We shall first examine what information does to a probability model.

Information removes impossible cases

Suppose a die is rolled once. Before any information is given, the natural sample space is

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

If the die is fair, all six elementary outcomes are equally likely.

Now imagine that someone tells us:

the result is even.

This information changes what is still possible. The outcomes

$$1, 3, 5$$

are no longer compatible with the information. The only outcomes that remain possible are

$$2, 4, 6.$$

Thus we no longer reason inside the whole set Ω. We reason inside the smaller event

$$E = \{2, 4, 6\}.$$

If we now ask

what is the probability that the result is greater than 3?

we must not count favourable outcomes inside the original six outcomes. The information has already told us that the result is even. Therefore the relevant possibilities are only

$$2, 4, 6.$$

Among these, the outcomes greater than 3 are

$$4, 6.$$

So, after receiving the information that the result is even, the probability of "greater than 3" is obtained by comparing two favourable outcomes with the three outcomes still possible.

Remark

The old sample space has not disappeared from the model. It still explains what the experiment was. But for the new question, after the information is known, the active space of reasoning is the event compatible with that information.

A question may change its answer after information is given

Before any information is given, the event

$$A = \{4, 5, 6\}$$

that the die result is greater than 3 has three favourable outcomes out of six. In the fair die model, this gives probability $1/2$.

After the information

$$E = \{2, 4, 6\}$$

is given, the same verbal question is evaluated in a smaller space. The relevant outcomes are no longer

$$1, 2, 3, 4, 5, 6,$$

but only

$$2, 4, 6.$$

Inside that restricted space, the event "greater than 3" corresponds to

$$A \cap E = \{4, 6\}.$$

Thus the answer changes. The event A itself did not change as a subset of Ω. What changed is the set of outcomes that are still considered possible.

This is the basic phenomenon of conditioning: information changes the reference space.

Information as restriction to an event

In probability, information usually has the form

$$\text{we know that } B \text{ occurred.}$$

The event B then becomes the new space of reasoning. Outcomes outside B are no longer possible under the information.

If we are interested in another event A, we do not ask how much of Ω belongs to A. We ask how much of the still possible region B also belongs to A. The relevant part is therefore

$$A \cap B.$$

The geometry of the situation is:

$$\begin{aligned} \Omega \quad & \text{is the original universe,} \\ B \quad & \text{is the information now known,} \\ A \cap B \quad & \text{is the part of the information region where } A \text{ also holds.} \end{aligned}$$

Definition

To **condition on information** means to restrict attention to the outcomes compatible with that information and then measure the event of interest inside that restricted region.

This is not yet a computational formula. It is the idea that the formula must express.

Two dice: information about the sum

Consider two fair dice, with order recorded. The original sample space is

$$\Omega = \{(i,j) : i,j \in \{1,2,3,4,5,6\}\}.$$

It contains 36 ordered pairs.

Let

$$B = \{\text{the sum is } 7\}.$$

Explicitly,

$$B = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}.$$

Suppose this information is given: the sum is 7. Then the six ordered pairs above are the only outcomes still compatible with what we know.

Now ask:

$$\text{what is the probability that the first die shows 5?}$$

Let

$$A = \{\text{the first die shows } 5\}.$$

Inside the whole sample space, A contains six outcomes:

$$(5,1),(5,2),(5,3),(5,4),(5,5),(5,6).$$

But most of these are irrelevant once we know that the sum is 7. Inside the restricted space B, the only outcome with first coordinate 5 is

$$(5,2).$$

Thus, after the information is known, there is one favourable outcome among the six outcomes still possible.

This example shows why one must be careful with the phrase "the probability of A". Probability relative to what information? Before knowing the sum, the question is evaluated in Ω. After knowing the sum, the question is evaluated in B.

The event is not replaced by the information

A common misunderstanding is to think that once B is known, the event A is replaced by B. That is not correct. The event of interest remains A. The information is B. The favourable outcomes are those where both things are true:

$$A \cap B.$$

In the two-dice example:

- B says that the sum is 7;
- A says that the first die is 5;
- $A \cap B$ says that the first die is 5 and the sum is 7.

The information gives the new reference space. The intersection gives the part of that reference space where the target event occurs.

Remark

Conditioning always involves two roles: the event being asked about and the information being conditioned on. Confusing these roles is one of the most common sources of wrong answers.

Cards: the same card viewed after information

Draw one card from a standard deck. Suppose the exact card is recorded, so the sample space has 52 elementary outcomes.

Let

$$A = \{\text{the card is an ace}\}.$$

Before any additional information is given, there are 4 aces among 52 cards.

Now suppose we are told:

$$B = \{\text{the card is a heart}\}.$$

The information reduces the relevant possibilities to the 13 hearts. Inside this restricted set, only one card is an ace: the ace of hearts. Therefore, once we know that the card is a heart, the event "the card is an ace" is evaluated inside the set of hearts.

Now change the information. Suppose instead we are told:

$$C = \{\text{the card is a face card}\}.$$

The relevant possibilities are now the face cards. Inside that restricted set, there are no aces, because aces are not face cards in the usual convention. Thus the same event A, "the card is an ace", behaves differently under different information.

The event A did not change. The information changed the reference space.

Information may or may not change the probability

New information restricts the space of reasoning, but it does not always change the numerical probability of the event of interest.

For example, draw one card from a standard deck. Let

$$A = \{\text{the card is an ace}\},$$

and let

$$B = \{\text{the card is a heart}\}.$$

Knowing that the card is a heart leaves 13 possible cards, one of which is an ace. The probability of being an ace inside the hearts is therefore the same as the original proportion of aces in the whole deck:

$$\text{one ace among thirteen hearts, just as four aces among fifty-two cards.}$$

In this case the information "heart" does not change the probability of "ace".

But if the information is

$$C = \{\text{the card is a face card}\},$$

then the probability of being an ace becomes zero inside the restricted space. This information does change the probability.

This observation prepares the idea of independence. Two events are independent when learning that one occurred does not change the probability of the other. But before defining independence formally, we must first understand what it means to update the space of reasoning.

Renormalization: the whole information region becomes certain

When the information B is given, the event B itself becomes certain in the new situation. After being told that the die result is even, the statement "the die result is even" has probability 1 relative to the information. After being told that the card is a heart, the statement "the card is a heart" has probability 1 relative to the information.

This means that probabilities inside the restricted region must be rescaled so that the whole information region has total probability 1. In a uniform finite model, this rescaling is easy to see. If B contains several equally likely outcomes, then after we learn B, those outcomes form the new complete list of possibilities.

Example

A fair die is rolled and we learn that the result is even. The restricted space is

$$B = \{2, 4, 6\}.$$

Inside this new space, the three outcomes $2, 4, 6$ are treated as the complete possibilities. Therefore the event B itself has probability 1 relative to this information, and each of the three remaining outcomes has probability $1/3$ in the restricted uniform model.

In non-uniform models, the same idea holds, but the remaining probabilities must be rescaled proportionally. The original weights outside B are discarded because they correspond to outcomes no longer compatible with the information. The weights inside B are then adjusted so that their total becomes 1.

This rescaling principle is the reason the formal definition of conditional probability has the structure it has. The numerator must measure the part of the information region where the target event occurs. The denominator must measure the whole information region, because that whole region becomes the new unit of probability.

Why the information must have positive probability

Conditioning on information only makes sense if the information can occur in the model. If $P(B) = 0$, then the instruction "reason inside B" cannot be implemented by rescaling probabilities inside B, because there is no positive probability mass to rescale.

In finite elementary examples this is easy to understand. Suppose a standard die is rolled and someone says:

the result is 8.

This statement corresponds to the empty event in the die model $\Omega = \{1, 2, 3, 4, 5, 6\}$. There are no outcomes compatible with it. It cannot become a new space of reasoning.

More generally, conditional probability is defined only when the conditioning event has positive probability. This condition is not a technical decoration. It says that the information must leave a meaningful region in which probability can be redistributed.

The conceptual picture

The logic of conditioning can now be summarized as follows.

1. We begin with a probability model on Ω.
2. We receive information that an event B occurred.
3. Outcomes outside B are no longer compatible with the information.
4. The event B becomes the new reference space.

5. To study another event A, we look at the part of B where A is also true, namely $A \cap B$.
6. Probabilities inside B are rescaled so that B has total probability 1 in the new situation.

Theorem

The guiding principle of conditional probability is:

probability after information is probability inside the event described by that information.

The formal definition in the next section will turn this principle into a precise rule.

The important point is that the formula will not be arbitrary. It must express the geometry we have already seen: restrict to the information event, keep only the part where the target event also occurs, and renormalize the information region so that it becomes the new whole space.

5.2 Conditional probability

The previous section explained the idea before the formula: when information is received, the relevant space of reasoning is restricted. If we know that an event B occurred, then outcomes outside B are no longer compatible with what we know. If we then ask about another event A, the favourable outcomes are those in

$$A \cap B.$$

We now turn this idea into a precise definition. The definition should not be read as an arbitrary rule. It is the mathematical expression of two decisions: first restrict to the information event, then rescale so that the information event becomes the new whole space.

The finite uniform case first

Let Ω be a finite sample space in which all elementary outcomes are equally likely. Suppose we are told that an event B occurred, and assume B is not empty. After this information, the possible outcomes are the elements of B.

If we ask for the probability of A after knowing B, then only the outcomes in B matter. Among these, the favourable outcomes are precisely the elements of

$$A \cap B.$$

Thus, in the finite uniform case, the natural value is

$$\frac{|A \cap B|}{|B|}.$$

This expression is not guessed. It is the ordinary uniform probability formula, but applied in the restricted space B instead of the original space Ω.

Example

A fair die is rolled. Let

$$A = \{4, 5, 6\}$$

be the event that the result is greater than 3, and let

$$B = \{2, 4, 6\}$$

be the event that the result is even. If we know B, then the relevant space is $\{2, 4, 6\}$.

Inside this space, the event A occurs for 4 and 6. Therefore the probability of A after knowing B is

$$\frac{2}{3}.$$

This is the ratio $|A \cap B|/|B|$, because

$$A \cap B = \{4, 6\}.$$

From counting to probability mass

The expression

$$\frac{|A \cap B|}{|B|}$$

works only when the restricted elementary outcomes are equally likely. In a general probability model, outcomes may have different probabilities. Then counting outcomes is not enough. We must measure probability mass.

The information event B has probability $P(B)$. This is the total mass of all outcomes compatible with the information. The event $A \cap B$ has probability $P(A \cap B)$. This is the part of that mass for which the event A also occurs.

After learning B, the whole region B becomes the new reference space. It must therefore have total probability 1 in the updated situation. The part of B where A occurs should occupy the same relative share that it had inside B before rescaling. That relative share is

$$\frac{P(A \cap B)}{P(B)}.$$

This gives the definition.

Definition

Let A and B be events in a probability space, and assume

$$P(B) > 0.$$

The **conditional probability of A given B** is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

The condition $P(B) > 0$ is essential. If the information event has probability zero, then it cannot be made into a new probability-one reference region by ordinary rescaling.

Why the denominator is $P(B)$

The denominator is not a technical detail. It represents the whole amount of probability still compatible with the information. Once we know B, the event B must become certain in the new calculation:

$$P(B \mid B) = 1.$$

The definition gives exactly this:

$$P(B \mid B) = \frac{P(B \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1.$$

Similarly, an event impossible under the information should have conditional probability zero. If $A \cap B = \varnothing$, then

$$P(A \mid B) = \frac{P(\varnothing)}{P(B)} = 0.$$

This matches the idea that, once B is known, an event disjoint from B cannot occur.

Conditional probability as a new probability assignment

For a fixed event B with $P(B) > 0$, the rule

$$A \mapsto P(A \mid B)$$

defines a new probability assignment on the original events, but interpreted under the information B.

It has the usual properties. First,

$$P(A \mid B) \geq 0,$$

because probabilities are non-negative. Second,

$$P(\Omega \mid B) = \frac{P(\Omega \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1.$$

Third, if A_1 and A_2 are disjoint, then

$$P((A_1 \cup A_2) \mid B) = \frac{P((A_1 \cup A_2) \cap B)}{P(B)}.$$

Since

$$(A_1 \cup A_2) \cap B = (A_1 \cap B) \cup (A_2 \cap B),$$

and the two pieces on the right are disjoint, additivity gives

$$P((A_1 \cup A_2) \mid B) = \frac{P(A_1 \cap B) + P(A_2 \cap B)}{P(B)}.$$

Therefore

$$P((A_1 \cup A_2) \mid B) = P(A_1 \mid B) + P(A_2 \mid B).$$

This confirms that conditional probability is not a separate kind of arithmetic. It is ordinary probability after changing the reference space.

Multiplication rule

The definition can be rewritten as

$$P(A \cap B) = P(A \mid B)P(B).$$

This is called the multiplication rule. It says: to get the probability that both A and B occur, first measure the probability of the information region B, and then multiply by the probability that A occurs inside that region.

The same intersection can also be read in the opposite order:

$$P(A \cap B) = P(B \mid A)P(A),$$

provided $P(A) > 0$.

Thus

$$P(A \mid B)P(B) = P(B \mid A)P(A).$$

This identity will later become the basis of Bayes' theorem.

Remark

The multiplication rule is not an additional assumption. It is the definition of conditional probability rearranged. Its usefulness comes from the fact that many multi-stage situations are naturally described by first entering one event and then asking for another event inside it.

Example: drawing two cards without replacement

Draw two cards from a standard deck without replacement, and record the ordered sequence. Let

$$A = \{\text{the first card is an ace}\},$$

and

$$B = \{\text{the second card is an ace}\}.$$

We want to understand the probability that both cards are aces.

First,

$$P(A) = \frac{4}{52}.$$

If A has occurred, then one ace has already been removed. There are now 51 cards left, of which 3 are aces. Therefore

$$P(B \mid A) = \frac{3}{51}.$$

By the multiplication rule,

$$P(A \cap B) = P(A)P(B \mid A) = \frac{4}{52} \cdot \frac{3}{51}.$$

This calculation is not merely a formula. It follows the physical structure of the experiment: first card, then second card after the first card has changed the deck.

Example: medical-style testing without Bayes yet

Suppose a population is divided into two groups: people who have a certain condition and people who do not. Let

$$C = \{\text{the person has the condition}\},$$

and let

$$T = \{\text{the test result is positive}\}.$$

The conditional probability

$$P(T \mid C)$$

means: among people with the condition, what proportion test positive?

The conditional probability

$$P(C \mid T)$$

means: among people with a positive test result, what proportion have the condition?

These are different questions. They use different reference spaces. The first uses C as the information event. The second uses T as the information event. There is no reason for the two numbers to be equal.

This distinction is one of the main reasons conditional probability must be introduced carefully. The vertical bar in $P(A \mid B)$ is not punctuation. It separates the event being measured from the information that defines the new space of reasoning.

Conditional probabilities in tables

Conditional probability can often be read from a table. Suppose a group of students is classified by whether they passed an exam and whether they attended review sessions:

	passed	did not pass	total
attended	36	9	45
did not attend	24	31	55
total	60	40	100

Let

$$A = \{\text{student passed}\},$$

and

$$B = \{\text{student attended}\}.$$

Then $P(A \mid B)$ is not computed from all 100 students. The information B restricts attention to the 45 students who attended. Among them, 36 passed. Thus

$$P(A \mid B) = \frac{36}{45}.$$

By contrast,

$$P(B \mid A) = \frac{36}{60},$$

because now the information is that the student passed, and among the 60 students who passed, 36 attended. Again, the two conditional probabilities answer different questions.

What this definition teaches

The definition

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

should be remembered through its meaning:

- B is the information that becomes the new reference space;
- $A \cap B$ is the part of that reference space where the event of interest also occurs;
- division by $P(B)$ rescales the information region so that it has total probability 1.

Theorem

Conditional probability is ordinary probability after the space of reasoning has been restricted by information. The formula records this restriction and the necessary rescaling.

5.3 Independence and dependence

Conditional probability lets us ask how the probability of one event changes after another event is known. The next natural question is therefore:

when does information change the probability, and when does it not?

This is the conceptual origin of independence. Independence is not first of all a multiplication formula. It is a statement that learning one event occurred does not change the probability of another event.

Information that changes probability

Draw one card from a standard deck. Let

$$A = \{\text{the card is an ace}\}.$$

Before any information is given,

$$P(A) = \frac{4}{52} = \frac{1}{13}.$$

Now suppose we are told that the card is a face card. Let

$$B = \{\text{the card is a face card}\}.$$

Inside the face cards there are no aces, using the standard convention that face cards are jacks, queens, and kings. Therefore

$$P(A \mid B) = 0.$$

The information B changes the probability of A. It changes it from $1/13$ to 0. In this case, the events are dependent.

Information that does not change probability

Now let

$$C = \{\text{the card is a heart}\}.$$

If we are told that the card is a heart, the relevant space consists of the 13 hearts. Among them, exactly one is an ace. Hence

$$P(A \mid C) = \frac{1}{13}.$$

But this is the same as the original probability $P(A)$. Knowing that the card is a heart does not change the probability that it is an ace.

This is the basic meaning of independence:

> the information occurred, but it did not change the probability of the event of interest.

Definition

Events A and B, with $P(B) > 0$, are **independent in the conditional sense** if

$$P(A \mid B) = P(A).$$

They are **dependent** if knowing B changes the probability of A, that is, if

$$P(A \mid B) \neq P(A).$$

This definition expresses the intuitive meaning directly. It says that B brings no probabilistic information about A.

The multiplication form of independence

The conditional definition is conceptually clear, but it requires $P(B) > 0$. Using the definition of conditional probability, the equality

$$P(A \mid B) = P(A)$$

becomes

$$\frac{P(A \cap B)}{P(B)} = P(A).$$

Multiplying both sides by $P(B)$, we obtain

$$P(A \cap B) = P(A)P(B).$$

This gives the standard symmetric definition.

Definition

Events A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

If this equality fails, the events are **dependent**.

This form is symmetric in A and B. It makes clear that if A gives no information about B, then B gives no information about A, provided the relevant conditional probabilities are defined.

Remark

The product formula should not be treated as the first meaning of independence. It is the algebraic form of the idea that conditioning on one event does not change the probability of the other.

Example: two coin tosses

Toss a fair coin twice and record the ordered sequence:

$$\Omega = \{HH, HT, TH, TT\}.$$

Let

$$A = \{\text{the first toss is heads}\} = \{HH, HT\},$$

and

$$B = \{\text{the second toss is heads}\} = \{HH, TH\}.$$

Then

$$P(A) = \frac{1}{2}, \qquad P(B) = \frac{1}{2}.$$

The intersection is

$$A \cap B = \{HH\},$$

so

$$P(A \cap B) = \frac{1}{4}.$$

Since

$$P(A)P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4},$$

we have

$$P(A \cap B) = P(A)P(B).$$

Thus A and B are independent.

The conditional interpretation says the same thing. If we know the second toss is heads, the remaining possible sequences are

$$HH, TH.$$

Among these, exactly one has first toss heads. Hence

$$P(A \mid B) = \frac{1}{2} = P(A).$$

The information about the second toss does not change the probability of the first toss.

Example: cards and dependence

Draw one card from a deck. Let

$$A = \{\text{the card is an ace}\},$$

and

$$B = \{\text{the card is a face card}\}.$$

Then

$$A \cap B = \varnothing,$$

so

$$P(A \cap B) = 0.$$

But

$$P(A)P(B) = \frac{4}{52} \cdot \frac{12}{52} > 0.$$

Therefore

$$P(A \cap B) \neq P(A)P(B).$$

The events are dependent. Knowing that the card is a face card makes the event “ace” impossible.

Disjointness is not independence

Students often confuse disjoint events with independent events. These ideas are very different.

Events A and B are disjoint if

$$A \cap B = \varnothing.$$

This means they cannot occur together.

Events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

This means that knowing one occurred does not change the probability of the other.

If A and B are disjoint and both have positive probability, then they cannot be independent. Indeed,

$$P(A \cap B) = 0,$$

but

$$P(A)P(B) > 0.$$

So the equality required for independence fails.

Remark

If two positive-probability events are disjoint, then learning that one occurred makes the other impossible. That is the strongest possible form of dependence, not independence.

Independence is not causation

Dependence means that probabilities change under information. It does not by itself identify a causal mechanism. If

$$P(A \mid B) \neq P(A),$$

then B is informative about A, but this does not automatically mean that B causes A.

For example, suppose A is the event that a randomly selected student passes an exam and B is the event that the student attended review sessions. If $P(A \mid B) > P(A)$, then attendance is associated with a higher pass probability in the model or data. But this observation alone does not prove that attendance caused the result. There may be other factors, such as preparation, motivation, or prior knowledge.

Probability detects changes of likelihood under information. Causation requires additional assumptions and is a separate question.

Independence of repeated trials as a modelling assumption

When we say that repeated coin tosses are independent, we are making a modelling assumption. For two tosses, this assumption says, for example, that knowing the first toss was heads does not change the probability that the second toss is heads.

If $P(H) = p$ for each toss and the tosses are independent, then

$$P(HH) = p^2,$$

$$\begin{aligned} P(HT) &= p(1-p), \\ P(TH) &= (1-p)p, \\ P(TT) &= (1-p)^2. \end{aligned}$$

Without the independence assumption, we cannot automatically multiply the single-toss probabilities. The second toss might depend on the first in some artificial mechanism. Independence is not guaranteed by notation such as $\{H,T\}^2$; it must be part of the probability assignment.

Example

Suppose a device produces two symbols, each either H or T, and the sample space is $\{HH, HT, TH, TT\}$. If the device always repeats the first symbol, then only HH and TT have positive probability. The sample space still uses two positions, but the two positions are not independent.

More than two events

For several events, independence becomes more subtle. It is not enough that each pair behaves independently. For three events A, B, C, full mutual independence requires the product rule for every relevant intersection:

$$\begin{aligned} P(A \cap B) &= P(A)P(B), \\ P(A \cap C) &= P(A)P(C), \\ P(B \cap C) &= P(B)P(C), \end{aligned}$$

and also

$$P(A \cap B \cap C) = P(A)P(B)P(C).$$

At this stage, the main point is conceptual: independence means that information from some events does not change probabilities of the others. For many events, this must hold consistently across combinations of events.

What independence teaches

Independence is a statement about information. It says that the occurrence of one event does not alter the probability assigned to another event. The product formula is important because it gives an algebraic test, but the meaning remains conditional:

$$\text{knowing } B \text{ does not change the probability of } A.$$

Theorem

For events of positive probability, the following ideas are equivalent:

$$P(A \mid B) = P(A),$$

$$P(B \mid A) = P(B),$$

$$P(A \cap B) = P(A)P(B).$$

Each expresses the same idea: the events do not provide probabilistic information about each other.

5.4 Partitions and the law of total probability

Conditional probability becomes especially useful when a problem can be split into cases. Many probability questions have the following structure: we want the probability of an event A, but the mechanism that produces A depends on which one of several possible background cases occurred.

The mathematical language for such a situation is a partition.

Splitting the sample space into cases

Suppose the sample space Ω is split into events

$$B_1, B_2, \ldots, B_n.$$

These events should represent mutually exclusive and exhaustive cases. Mutually exclusive means that no two cases can occur together:

$$B_i \cap B_j = \varnothing \qquad \text{when } i \neq j.$$

Exhaustive means that every possible outcome belongs to one of the cases:

$$B_1 \cup B_2 \cup \cdots \cup B_n = \Omega.$$

Definition

A finite collection of events $B_1, \ldots, B_n$ is a **partition** of Ω if the events are pairwise disjoint and their union is Ω. In applications of conditional probability, we usually also require

$$P(B_i) > 0$$

for each case that appears in a conditional probability.

A partition is a disciplined way to say: exactly one of these cases occurs.

Example

If a person is selected from a population, the events

$$B_1 = \{\text{person is in group } 1\}, \quad B_2 = \{\text{person is in group } 2\}, \quad B_3 = \{\text{person is in group } 3\}$$

form a partition if every person belongs to exactly one of the three groups.

How an event is split by a partition

Let A be any event. If $B_1, \dots, B_n$ is a partition of Ω, then A can be split according to the cases:

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \cdots \cup (A \cap B_n).$$

These pieces are disjoint because the B_i's are disjoint. Thus the probability of A is the sum of the probabilities of the pieces:

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \cdots + P(A \cap B_n).$$

This identity is just additivity applied after decomposing A into disjoint parts.

Replacing intersections by conditional probabilities

For each case B_i, the multiplication rule gives

$$P(A \cap B_i) = P(A \mid B_i)P(B_i).$$

Substituting this into the previous sum gives

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + \cdots + P(A \mid B_n)P(B_n).$$

Equivalently,

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

Theorem

Law of total probability. If $B_1, \dots, B_n$ is a partition of Ω and $P(B_i) > 0$, then for every event A,

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

The formula says that the total probability of A is obtained by looking at all possible cases. In each case, we multiply the probability of the case by the probability of A inside that case, and then add the contributions.

Remark

The law of total probability is not a new mysterious rule. It is additivity over a partition, with each piece $P(A \cap B_i)$ rewritten as $P(A \mid B_i)P(B_i)$.

Weighted averages of conditional probabilities

The expression

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i)$$

can be read as a weighted average of the conditional probabilities

$$P(A \mid B_1), \dots, P(A \mid B_n).$$

The weights are

$$P(B_1), \dots, P(B_n).$$

They are non-negative and add to one because the B_i's form a partition:

$$P(B_1) + \cdots + P(B_n) = 1.$$

This interpretation is useful. If A is likely in a case that rarely occurs, its contribution to the total probability may still be small. If A is only moderately likely in a case that occurs often, its contribution may be large.

Example: choosing one of two coins

Suppose a box contains two coins. Coin 1 is fair:

$$P(H \mid B_1) = \frac{1}{2}.$$

Coin 2 is biased toward heads:

$$P(H \mid B_2) = \frac{3}{4}.$$

A coin is chosen at random, with

$$P(B_1) = P(B_2) = \frac{1}{2},$$

and then the chosen coin is tossed once. What is the probability of heads?

Let

$$A = \{\text{heads occurs}\}.$$

The events B_1 and B_2 form a partition of the background mechanism: we either chose coin 1 or coin 2. The law of total probability gives

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2).$$

Thus

$$P(A) = \frac{1}{2} \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2} = \frac{1}{4} + \frac{3}{8} = \frac{5}{8}.$$

The answer is between 1/2 and 3/4, because before the toss we do not know which coin was chosen. The total probability is the average of the head probabilities of the two coins, weighted by the probabilities of choosing them.

Example: urn chosen before drawing

Suppose one urn is selected first, and then one object is drawn from that urn. Urn 1 contains 3 red objects and 1 blue object. Urn 2 contains 1 red object and 4 blue objects. The probability of choosing urn 1 is 2/3, and the probability of choosing urn 2 is 1/3.

Let

$$A = \{\text{the drawn object is red}\},$$

$$B_1 = \{\text{urn 1 was chosen}\}, \qquad B_2 = \{\text{urn 2 was chosen}\}.$$

Then

$$P(A \mid B_1) = \frac{3}{4}, \qquad P(A \mid B_2) = \frac{1}{5}.$$

The cases B_1 and B_2 form a partition. Therefore

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2).$$

Substituting the values,

$$P(A) = \frac{3}{4} \cdot \frac{2}{3} + \frac{1}{5} \cdot \frac{1}{3}.$$

Hence

$$P(A) = \frac{1}{2} + \frac{1}{15} = \frac{17}{30}.$$

The result is not obtained by averaging the contents of the two urns equally, because the urns are not chosen equally. The weights 2/3 and 1/3 matter.

Example: data table

Suppose a population consists of two departments. Department 1 contains 40% of the population, and department 2 contains 60%. In department 1, 70% of people use a certain system. In department 2, 30% use it.

Let

$$A = \{\text{the selected person uses the system}\},$$
$$B_1 = \{\text{the selected person is in department 1}\},$$
$$B_2 = \{\text{the selected person is in department 2}\}.$$

Then

$$P(B_1) = 0.4, \qquad P(B_2) = 0.6,$$
$$P(A \mid B_1) = 0.7, \qquad P(A \mid B_2) = 0.3.$$

Therefore

$$P(A) = 0.7 \cdot 0.4 + 0.3 \cdot 0.6 = 0.28 + 0.18 = 0.46.$$

The overall usage rate is 46%. It is a weighted average of the department usage rates, weighted by department sizes.

Choosing the right partition

The law of total probability is useful only after the cases have been chosen well. A good partition should satisfy three conditions.

First, the cases must be mutually exclusive. If two cases can occur together, then adding their contributions may double-count outcomes.

Second, the cases must be exhaustive. If some outcomes are not covered by any case, then the sum will miss part of the probability.

Third, the conditional probabilities inside the cases should be meaningful and usually easier to understand than the original probability.

Example

For a test problem, a natural partition may be

$$B_1 = \{\text{person has the condition}\}, \qquad B_2 = \{\text{person does not have the condition}\}.$$

These two events are disjoint and exhaustive. They also match how test accuracy is usually described: sensitivity gives $P(\text{positive} \mid B_1)$, and false positive rate gives $P(\text{positive} \mid B_2)$.

Common mistake: forgetting the weights

A common mistake is to average conditional probabilities without using the case probabilities. For example, if

$$P(A \mid B_1) = 0.9, \qquad P(A \mid B_2) = 0.1,$$

one might be tempted to say that $P(A) = 0.5$. This is correct only if

$$P(B_1) = P(B_2) = 0.5.$$

If instead

$$P(B_1) = 0.1, \qquad P(B_2) = 0.9,$$

then

$$P(A) = 0.9 \cdot 0.1 + 0.1 \cdot 0.9 = 0.18.$$

The rare high-probability case contributes less than the common low-probability case.

What the law of total probability teaches

The law of total probability teaches how to assemble a probability from cases. The event A is decomposed into disjoint pieces $A \cap B_i$. Each piece is measured by first entering case B_i, then measuring A within that case.

Theorem

To use total probability, ask:

Which mutually exclusive and exhaustive cases can produce the event?

Then add the case contributions:

$$\text{probability of case} \times \text{probability of event inside case.}$$

This prepares the next idea. If total probability lets us compute the probability of an observation from possible causes or cases, Bayes' theorem lets us reverse the question: after the observation is known, how likely is each case?

5.5 Bayes' theorem

The law of total probability lets us compute the probability of an observation by splitting the world into possible cases. Bayes' theorem answers the reverse question. After an observation is known, how should we update the probabilities of the possible cases?

This is not a separate trick. It is conditional probability applied in the reverse direction.

The basic reversal problem

Suppose B is a possible background case and A is an observed event. We may know or be able to model

$$P(A \mid B),$$

the probability of observing A if case B is true. But after observing A, we may want

$$P(B \mid A),$$

the probability that case B is true given that A was observed.

These two probabilities are not the same. They use different reference spaces. The first reasons inside B. The second reasons inside A.

From the multiplication rule,

$$P(A \cap B) = P(A \mid B)P(B).$$

Also,

$$P(A \cap B) = P(B \mid A)P(A).$$

Since both expressions describe the same intersection, we get

$$P(B \mid A)P(A) = P(A \mid B)P(B).$$

If $P(A) > 0$, then

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}.$$

Theorem

Bayes' theorem, two-event form. If $P(A) > 0$ and $P(B) > 0$, then

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}.$$

This formula says that the updated probability of B after observing A is proportional to two quantities:

- how plausible B was before observing A, namely $P(B)$;
- how well B explains the observation A, namely $P(A \mid B)$.

Why the denominator is often expanded

In many applications, $P(A)$ is not known directly. Instead, the observation A can occur under several possible cases. If $B_1, \ldots, B_n$ form a partition of Ω, then the law of total probability gives

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

Substituting this into Bayes' theorem gives

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}.$$

Theorem

Bayes' theorem over a partition. If $B_1, \ldots, B_n$ is a partition of Ω, $P(B_i) > 0$, and $P(A) > 0$, then

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}.$$

The denominator is not an arbitrary normalizing term. It is the total probability of the observation A, obtained by adding all the ways in which A can occur through the cases B_i.

Prior, likelihood, posterior

Bayes' theorem is often described using three words.

The number

$$P(B_j)$$

is the **prior probability** of case B_j. It is the probability before observing A.

The number

$$P(A \mid B_j)$$

is the **likelihood** of the observation under case B_j. It measures how compatible the observation is with that case.

The number

$$P(B_j \mid A)$$

is the **posterior probability**. It is the updated probability of the case after observing A.

The logic is:

$$\text{prior} \times \text{likelihood} \quad \longrightarrow \quad \text{posterior after normalization.}$$

The normalization is necessary because the posterior probabilities over all cases must add to one.

Example: choosing an urn after observing a colour

Suppose one of two urns is chosen. Urn 1 is chosen with probability 2/3, and urn 2 is chosen with probability 1/3. Urn 1 contains 3 red objects and 1 blue object. Urn 2 contains 1 red object and 4 blue objects. One object is drawn from the chosen urn, and it is red. Which urn is now more likely to have been chosen?

Let

$$B_1 = \{\text{urn 1 was chosen}\}, \qquad B_2 = \{\text{urn 2 was chosen}\},$$

and

$$A = \{\text{the drawn object is red}\}.$$

We have

$$P(B_1) = \frac{2}{3}, \qquad P(B_2) = \frac{1}{3},$$
$$P(A \mid B_1) = \frac{3}{4}, \qquad P(A \mid B_2) = \frac{1}{5}.$$

First compute the total probability of drawing red:

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2).$$

Thus

$$P(A) = \frac{3}{4} \cdot \frac{2}{3} + \frac{1}{5} \cdot \frac{1}{3} = \frac{1}{2} + \frac{1}{15} = \frac{17}{30}.$$

Now Bayes' theorem gives

$$P(B_1 \mid A) = \frac{P(A \mid B_1)P(B_1)}{P(A)}.$$

Therefore

$$P(B_1 \mid A) = \frac{\frac{3}{4} \cdot \frac{2}{3}}{\frac{17}{30}} = \frac{\frac{1}{2}}{\frac{17}{30}} = \frac{15}{17}.$$

Similarly,

$$P(B_2 \mid A) = \frac{\frac{1}{5} \cdot \frac{1}{3}}{\frac{17}{30}} = \frac{\frac{1}{15}}{\frac{17}{30}} = \frac{2}{17}.$$

The posterior probabilities add to one:

$$\frac{15}{17} + \frac{2}{17} = 1.$$

The red observation strongly supports urn 1 because urn 1 was already more likely before the draw and also has a higher red proportion.

Example: diagnostic testing

Suppose a condition occurs in 1% of a population:

$$P(C) = 0.01, \qquad P(C^c) = 0.99.$$

A test has sensitivity 95%, meaning

$$P(T \mid C) = 0.95,$$

and false positive rate 5%, meaning

$$P(T \mid C^c) = 0.05.$$

Here T is the event that the test result is positive.

We want

$$P(C \mid T),$$

the probability that a person has the condition after receiving a positive test result.

By Bayes' theorem over the partition C, C^c,

$$P(C \mid T) = \frac{P(T \mid C)P(C)}{P(T \mid C)P(C) + P(T \mid C^c)P(C^c)}.$$

Substitute the values:

$$P(C \mid T) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.05 \cdot 0.99}.$$

The numerator is

$$0.0095.$$

The denominator is

$$0.0095 + 0.0495 = 0.059.$$

Therefore

$$P(C \mid T) = \frac{0.0095}{0.059} \approx 0.161.$$

So the probability is about 16.1%, not 95%.

This example is important because it shows the role of the base rate. A positive test result is evidence for the condition, but when the condition is rare, false positives may still be numerous among all positive results.

Natural frequency interpretation

The diagnostic example becomes clearer if we imagine 10000 people.

About 1% have the condition, so approximately

$$100$$

people have it, and

$$9900$$

do not.

Among the 100 people with the condition, the test is positive for about

$$0.95 \cdot 100 = 95$$

people.

Among the 9900 people without the condition, the test is positive for about

$$0.05 \cdot 9900 = 495$$

people.

So the total number of positive tests is about

$$95 + 495 = 590.$$

Among these, only 95 are true positives. Hence

$$P(C \mid T) \approx \frac{95}{590} \approx 0.161.$$

This is the same calculation as Bayes' theorem, but written as counts. It often makes the denominator easier to understand: the positive tests come from two sources, true positives and false positives.

Bayes' theorem as redistribution of probability over cases

Suppose $B_1, \dots, B_n$ are possible cases. Before observing A, the probability assigned to case B_i is $P(B_i)$. After observing A, each case receives support proportional to

$$P(A \mid B_i)P(B_i).$$

The product has a clear meaning:

$$P(A \mid B_i)P(B_i) = P(A \cap B_i).$$

It is the probability that case B_i occurs and produces the observation A.

After observing A, all posterior probabilities must add to one. Therefore we divide each contribution by the total probability of A:

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

This turns the contributions into posterior probabilities.

Common reversal error

The most common mistake is to confuse

$$P(A \mid B)$$

with

$$P(B \mid A).$$

For example, test sensitivity $P(T \mid C)$ is not the same as the probability of having the condition after a positive test $P(C \mid T)$. The first uses the condition group as the reference space. The second uses the positive-test group as the reference space.

Bayes' theorem exists precisely because these two directions are different but related through the intersection $A \cap B$.

Remark

Whenever a conditional probability appears, read the phrase after the vertical bar first. It tells you the reference space. Then read the event before the bar. It tells you what is being measured inside that reference space.

What Bayes' theorem teaches

Bayes' theorem teaches how probability is updated by evidence. It combines three ingredients:

- prior probabilities of the possible cases;
- likelihoods of the observation under those cases;
- normalization by the total probability of the observation.

Theorem

Bayes' theorem is conditional probability used backward. It converts

$$P(\text{observation} \mid \text{case})$$

into

$$P(\text{case} \mid \text{observation}),$$

using the prior probabilities of the cases and the total probability of the observation.

5.6 Diagnostic examples and common traps

Conditional probability is conceptually simple once the reference space is clear, but it is easy to make mistakes when the roles of events are not stated explicitly. This section collects common traps and shows how to diagnose them.

The guiding question is always:

what is the information, and what is being measured inside that information?

Trap 1: confusing $P(A \mid B)$ with $P(B \mid A)$

The conditional probability

$$P(A \mid B)$$

means that B is the information and A is the event measured inside that information. The conditional probability

$$P(B \mid A)$$

reverses these roles.

These two numbers are usually different.

Example

Let

$$A = \{\text{a person is a mathematics student}\},$$

and

$$B = \{\text{a person takes a probability course}\}.$$

The probability $P(B \mid A)$ asks: among mathematics students, what proportion take a probability course? The probability $P(A \mid B)$ asks: among students in the probability course, what proportion are mathematics students?
These are different reference groups, so the answers need not be the same.

Diagnostic question:

Which event is after the vertical bar?

That event is the new reference space.

Trap 2: ignoring base rates

Bayesian problems often involve a rare event and an imperfect observation. The observation may be strong evidence, but the prior probability still matters. Ignoring the prior probability is called ignoring the base rate.

Suppose a condition occurs in 1% of a population. A test is positive for 99% of people with the condition and for 5% of people without the condition. A positive test does not mean the person has probability 99% of having the condition. The number 99% is

$$P(\text{positive} \mid \text{condition}).$$

The desired probability after a positive test is

$$P(\text{condition} \mid \text{positive}).$$

These are different.

Diagnostic question:

How common was the case before the evidence was observed?

Bayes' theorem combines the prior probability with the likelihood of the observation.

Trap 3: treating disjoint events as independent

If two events cannot occur together, they are disjoint:

$$A \cap B = \varnothing.$$

If both have positive probability, then learning that one occurred makes the other impossible. Therefore they are dependent, not independent.

For example, in one die roll, let

$$A = \{\text{the result is even}\} = \{2, 4, 6\},$$

and

$$B = \{\text{the result is odd}\} = \{1, 3, 5\}.$$

The events are disjoint. If we know B, then the probability of A becomes zero:

$$P(A \mid B) = 0.$$

But before knowing B,

$$P(A) = \frac{1}{2}.$$

The information changes the probability, so the events are dependent.

Diagnostic question:

If one event occurs, does the other become impossible?

If yes, then positive-probability events are not independent.

Trap 4: assuming independence without justification

Independence is not created by notation. Writing a sample space as a product, such as

$$\{H, T\}^2$$

or

$$\{1, 2, 3, 4, 5, 6\}^2,$$

describes possible ordered outcomes. It does not by itself prove that the stages are independent.

Independence is a property of the probability assignment. It says that knowing one event occurred does not change the probability of another.

Example

A mechanism produces two symbols, each either H or T. The possible outputs are

$$HH, HT, TH, TT.$$

If the mechanism chooses the first symbol fairly and then always repeats it, the only outputs with positive probability are HH and TT. The two positions are not independent, even though the notation has two coordinates.

Diagnostic question:

Where in the model is independence stated or justified?

If it is not stated or implied by the mechanism, do not use product rules as if it were automatic.

Trap 5: using total probability without a partition

The law of total probability requires cases that are mutually exclusive and exhaustive. If the cases overlap, adding contributions may double-count. If the cases do not cover the whole sample space, the sum may miss part of the event.

Before writing

$$P(A) = \sum_i P(A \mid B_i)P(B_i),$$

check that the events B_i form a partition.

Diagnostic questions:

- Can two of the cases occur at the same time?
- Is every possible outcome covered by one of the cases?
- Do all conditioning events have positive probability?

Trap 6: forgetting the denominator in Bayes' theorem

In Bayes' theorem, the denominator is the total probability of the observation. It includes all ways the observation can occur. Omitting some cases changes the reference space and gives the wrong posterior probability.

For a partition $B_1, \ldots, B_n$, the denominator is

$$P(A) = \sum_i P(A \mid B_i)P(B_i).$$

It is not merely the likelihood under the case we are interested in.

Diagnostic question:

What are all possible sources of the observation?

The denominator must include them all.

Trap 7: treating a conditional probability as a causal statement

If

$$P(A \mid B) > P(A),$$

then B is informative about A. But this alone does not prove that B causes A. Conditional probability describes how probabilities change under information. Causal interpretation requires additional structure.

For example, if students who attend review sessions pass more often, then attendance is associated with passing. But from this probability statement alone we cannot conclude that attendance caused the higher pass rate. Students who attend may differ in other ways.

Diagnostic question:

Am I making a probability statement or a causal claim?

These are not the same kind of statement.

Trap 8: conditioning on an event of probability zero

The elementary formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

requires

$$P(B) > 0.$$

If $P(B) = 0$, the denominator vanishes and the formula is not defined.

In finite examples this often corresponds to impossible information. For a die roll, conditioning on the event "the result is 8" is not meaningful in the usual die model because that event is empty.

Diagnostic question:

Does the conditioning event have positive probability in this model?

A diagnostic workflow

When a conditional probability problem appears, use the following workflow.

1. Identify the original sample space or probability model.
2. Identify the information event: this is the event after the vertical bar.
3. Identify the target event: this is the event before the vertical bar.
4. Check that the information event has positive probability.
5. Replace the active space of reasoning by the information event.
6. Locate the target event inside that information event by taking the intersection.
7. Decide whether the problem needs direct conditioning, independence, total probability, or Bayes' theorem.

Theorem

Most mistakes with conditional probability are mistakes about reference spaces. Before calculating, say explicitly: "I am reasoning inside this event."

5.7 Final checklist

This chapter developed conditional probability as probability after information has changed the active space of reasoning. The central habit is to identify the reference space before calculating.

The checklist below summarizes the method.

Step 1: Identify the original model

Before conditioning, there must be a probability model. Identify:

- the sample space Ω;
- the elementary outcomes or relevant events;
- the probability assignment;
- whether the model is uniform or non-uniform.

Conditional probability does not replace model construction. It uses the model that has already been built.

Step 2: Identify the information event

In an expression

$$P(A \mid B),$$

the event after the vertical bar is the information event. It is the event that is assumed to have occurred.

Say explicitly:

$$\text{we are now reasoning inside } B.$$

Check that

$$P(B) > 0.$$

If $P(B) = 0$, the elementary conditional probability formula is not defined.

Step 3: Identify the target event

The event before the vertical bar is the event being measured. In

$$P(A \mid B),$$

the target event is A. The favourable part of the information region is

$$A \cap B.$$

Do not replace A by B. Do not reverse the order unless the problem asks for the reverse conditional probability.

Step 4: Use the definition when conditioning directly

If the problem asks directly for $P(A \mid B)$, use

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

In a finite uniform model this becomes

$$P(A \mid B) = \frac{|A \cap B|}{|B|}.$$

The denominator is the size or probability of the information region. The numerator is the part of that region where the target event also occurs.

Step 5: Use independence only when justified

Events A and B are independent when information about one does not change the probability of the other. Algebraically,

$$P(A \cap B) = P(A)P(B).$$

When probabilities are positive, this is equivalent to

$$P(A \mid B) = P(A)$$

and

$$P(B \mid A) = P(B).$$

Do not assume independence merely because there are two stages or two coordinates. Independence belongs to the probability assignment, not just to the notation.

Step 6: Use total probability when cases produce the event

If an event A may occur through several mutually exclusive and exhaustive cases $B_1, \dots, B_n$, use the law of total probability:

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

Before using it, check:

- the cases are disjoint;
- the cases cover the whole sample space;
- the conditional probabilities inside the cases are meaningful.

Read the formula as:

$$\text{total probability} = \sum \text{case probability} \times \text{probability inside case}.$$

Step 7: Use Bayes' theorem when reversing information

Use Bayes' theorem when the problem gives or suggests probabilities of the form

$$P(\text{observation} \mid \text{case})$$

but asks for

$$P(\text{case} \mid \text{observation}).$$

For a partition $B_1, \dots, B_n$,

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}.$$

Remember the roles:

- $P(B_j)$ is the prior probability of the case;
- $P(A \mid B_j)$ is the likelihood of the observation in that case;
- $P(B_j \mid A)$ is the posterior probability of the case after the observation.

Step 8: Interpret the result

After computing a conditional probability, translate it back into words.

For example,

$$P(A \mid B) = 0.7$$

means:

$$\text{among the outcomes compatible with } B, \text{ the event } A \text{ has probability } 0.7.$$

It does not mean that B was caused by A, nor that $P(B \mid A) = 0.7$.

Compact checklist

For any problem in this chapter, ask:

1. What is the original probability model?
2. What information has been given?
3. What event is after the vertical bar?
4. What event is before the vertical bar?
5. What is the intersection of the target event with the information event?
6. Is the conditioning event positive-probability?
7. Is independence stated or justified?
8. Is there a partition of cases?
9. Is the problem asking for a direct conditional probability or a reverse probability requiring Bayes' theorem?
10. What does the final number mean in words?

Theorem

The most important sentence in conditional probability is:

the event after the vertical bar is the space in which we now reason.

Once this is clear, the formulas become expressions of that change of reference space.

Chapter 6

Discrete Random Variables and Probability Laws

This chapter introduces random variables through the discrete case. A random variable is viewed as a function that extracts a numerical quantity from an elementary outcome and allows probabilistic phenomena to be studied through numerical characteristics.

The chapter develops discrete probability laws, probability mass functions, cumulative distribution functions, expectation, variance, moments, and the interpretation of these quantities. Classical discrete models are presented as mathematical descriptions of specific random mechanisms.

Chapter 7

Continuous Random Variables and Probability Laws

This chapter develops the continuous counterpart of discrete probability laws. The central conceptual change is that probability is no longer concentrated at individual points; instead, probabilities of intervals become the fundamental objects.

The chapter introduces probability density functions, cumulative distribution functions, integrals, expectation, variance, moments, quantiles, and the interpretation of probability as area under a density. Continuous models such as the uniform, exponential, normal, gamma, and beta distributions are treated as tools for modelling continuous random phenomena.

Chapter 8

Empirical Distributions and Descriptive Statistics

This chapter studies empirical distributions and descriptive statistics obtained from data. The concepts developed earlier for theoretical probability laws are now examined through observed datasets.

Frequency tables, histograms, empirical distribution functions, means, medians, variances, standard deviations, quantiles, and related descriptive measures are introduced as tools for describing and understanding data.

Chapter 9

Sampling and Distributions of Statistics

This chapter introduces the idea that statistics computed from samples are themselves random variables. Sample means, proportions, variances, medians, and other summaries acquire their own probability laws when sampling is repeated.

The purpose of the chapter is to prepare the transition from descriptive statistics to statistical inference through the study of sampling distributions.

Chapter 10

Limit Theorems and Statistical Inference

This chapter studies the regularities that emerge when sample sizes become large and experiments are repeated many times. The law of large numbers and the central limit theorem provide the mathematical foundation for statistical inference.

The chapter introduces normal approximations, standard errors, confidence intervals, and the basic principles of inference from data.